Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

The format of spatial memory in the hippocampus varies systematically with a species' visual ecology, but whether this reflects a general sensor–memory coupling is hard to test directly: animal sensor ecology is fixed within species, and frontier AI systems rarely co-train encoder and memory from scratch along a sensor axis. We test this instead in a silicon analogue: a navigation agent where only the visual sensor varies along an axis of encoder spatial bandwidth, with architecture, task, and training held fixed. Across this axis, agents reach near-identical task performance, yet the recurrent-state code for position differs in form. With a vision-limited encoder, position concentrates into a scene-invariant direction linearly readable from the recurrent state; with a rich encoder, it distributes into non-linear directions and partly offloads to the encoder. Linear readability shifts faster than total position information — a format shift rather than a total-information collapse. We frame this as an emergent *capacity allocation* between the visual encoder and the recurrent integration of GPS-sensor history. Behaviourally, memory transplants are asymmetric — rich-encoder policies fail to use bottleneck states, while bottleneck policies still drive themselves from rich-encoder ones — indicating policy-level consequences of the format differences. The pattern connects to the invariance–disentanglement reading of the information bottleneck and to the no-remapping vs. global-remapping axis of hippocampal coding under hidden-state inference. This positions sensor structure as a training-time lever on the format of cognitive maps in artificial agents, offering a content-level alternative to training-distribution accounts of vision–language spatial-reasoning failures. More broadly, this capacity allocation generates falsifiable predictions for transformer-based world models that share the encoder–memory pipeline, and suggests that bandwidth restriction may be cognition-enabling rather than merely compute-saving.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a pipeline: scale the encoder, and any downstream module benefits. Recent benchmarks complicate this picture. Vision–language models (VLMs) with state-of-the-art surface visual recognition still fail at fine-grained spatial reasoning (**?**). The default reading attributes this to a training-distribution gap. We pursue a structural reading instead: the encoder and the memory downstream of it are not separable modules but a co-trained system, and *the encoder's structure shapes what is linearly readable from the memory about space* — a content-level claim about the perception–memory interface.

Cognitive neuroscience offers concrete precedent: primate hippocampal cells respond to combinations of viewed location, head direction, and self-position (**??**), in contrast to the more place-pure coding documented in rodents with near-panoramic vision; flying bats globally remap their hippocampal code between vision and echolocation (**?**); and congenitally blind humans show larger hippocampal volume and supranormal non-visual spatial navigation (**?**). We treat these as motivation rather than

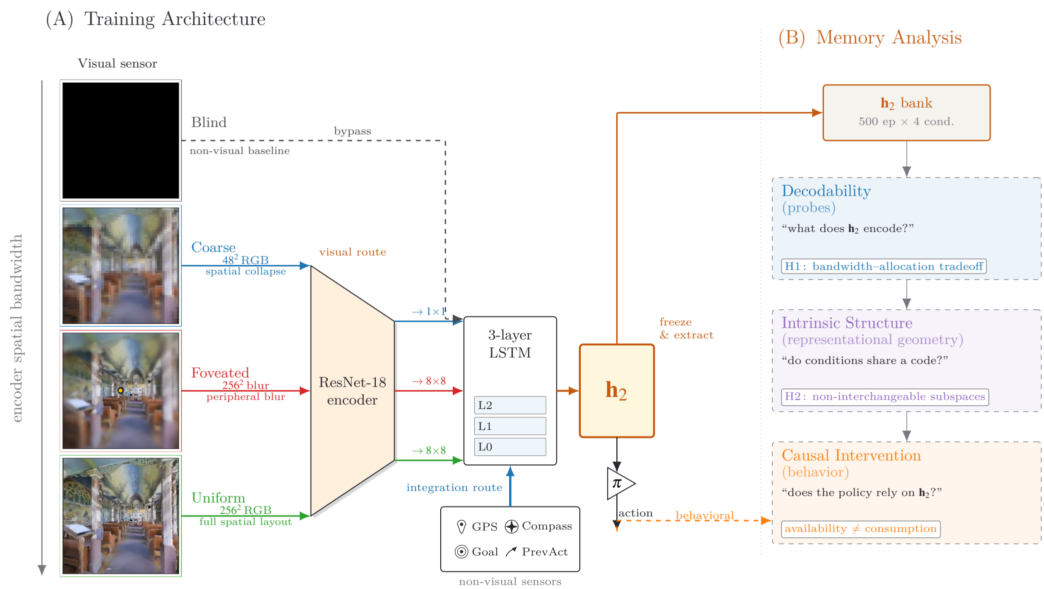

Figure 1: **Experimental setup and analysis pipeline.** *Row 1 (architecture):* four sensor conditions span an axis of encoder spatial bandwidth (*blind*, *coarse*, *foveated*, *uniform*; leftmost column), feeding a shared ResNet-18 encoder + 3-layer LSTM; *blind* bypasses the encoder. A log-polar foveation control (Appendix **??**) adds a 5th intermediate point on the same axis. *Row 2 (analysis):* three method blocks — decodability, intrinsic structure, causal intervention — read the frozen top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ to test three axes of one capacity allocation: *magnitude*, *format*, *consumption* (§**??**–**??**; synthesis in §**??**). Full setup in §**??**.

established parallel: whether sensor structure plays a similar shaping role in artificial navigation agents is a hypothesis to test in a controlled setting, not a parallel to demonstrate.

Map-like representations emerge in the recurrent state of deep-RL navigation agents without being designed in, even in agents with no vision at all (**???**).[1] This literature has typically reported one condition at a time — a fully sighted or a fully blind agent — leaving open how encoder structure interacts with what the memory builds. We adopt the same agent family as a *comparative* chassis (Figure **??**): identical encoder–memory–policy pipeline, task, and training; varying only the visual sensor across four conditions plus a log-polar control that span an axis of encoder spatial bandwidth (full definitions in §**??**).

Our central empirical finding is that *rich visual encoders reduce the linear-readable GPS-derived position signal in the policy-readable top-layer hidden state*: as encoder spatial bandwidth grows, top-layer linear GPS R^2 falls monotonically, even though task performance is near-identical across conditions. We frame this pattern as an emergent *capacity allocation* between two routes carrying spatial information — the visual encoder and the recurrent integration of GPS-sensor history — with the share each route carries tracking encoder spatial bandwidth. The principle predicts the allocation manifests along three orthogonal axes: a *magnitude* axis (how much linear position information lives in the recurrent state), a *format* axis (which subspace of the hidden state carries it across conditions), and a *consumption* axis (which route the policy actually uses for behaviour). For visual intelligence, this principle offers an architecture-side account complementing the dominant data-side account (**?**) of VLM spatial-reasoning failures (**?**): rich pretrained encoders may structurally divert capacity from the downstream substrate's spatial code, not merely fail to teach it — and conversely, bandwidth-restricting perception (foveation, log-polar sampling) may be cognition-*enabling* for downstream spatial tasks, not only compute-saving. For cognitive neuroscience, our setup is a controlled silicon

[1]"Cognitive map" in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept (**?**) are engaged by our LOSO scene-invariance test (§**??**, relational across rooms) and predictive-horizon analysis (Appendix **??**, SR-style future-state code in the sense of **?**); hierarchical scene clustering and topological reachability probes are deferred to follow-up work.

testbed that generates falsifiable predictions about how sensor-niche structure shapes spatial memory format (§**??**).

2 Related Work

This paper sits at the intersection of five lines of work: VLM spatial-reasoning failures, encoder–memory pipelines in embodied AI, foveated perception, emergent maps in deep-RL navigation, and cognitive-map / place-cell theory in neuroscience. We also draw on the interpretability methodology surveyed at the end of this section.

Spatial-reasoning failures in vision–language models Multimodal LLMs with strong visual recognition systematically fail at fine-grained spatial reasoning (**??**). Two diagnoses compete: a *data-side* account treats the gap as a training-distribution problem (**?**), and an *encoder-side* account treats it as structurally upstream of downstream training — probing studies show pretrained encoders with different objectives carry markedly different geometric content, with self-supervised encoders retaining depth and surface structure that vision–language-trained ones lack (**???**). Our paper provides a controlled analog of the encoder-side account in a recurrent navigation setting; whether the same mechanism transfers to attention-based memory in VLMs is a prediction, not a result.

Encoder–memory interface in modern embodied AI Recent world models (**??**) and embodied vision–language–action agents (**??**) share an architectural pattern: a pretrained visual encoder feeds an attention- or recurrence-based memory module read by a downstream policy. Practitioner choices — multi-encoder fusion (**?**), geometry-driven encoder selection (**?**), frozen-feature extraction (**??**) — already reflect that encoder content shapes downstream representations. Whether transformer-based navigators (**?**) integrate sensor history the way LSTM bottleneck conditions do here is an open question.

Foveated and bandwidth-constrained vision Peripheral-vision phenomenology (**??**) motivates deep-learning work applying eccentricity-dependent processing to neural networks, mostly under a compute or recognition prior (**??**). **?** are the closest precedent for treating foveation as content-shaping. **?** train PointGoalNav agents across a pixel-bandwidth spectrum without examining the recurrent state; **?** probe a foveated visual-search model (no embodied locomotion). We extend this line by asking how sensor bandwidth reshapes the format of downstream spatial memory in a recurrent navigation policy.

Emergent maps in deep-RL navigation Map-like structure emerges in the recurrent state of navigation agents trained under absent or spatially uniform perception (**???**). Self-supervised path-integration can produce multi-modular grid cells (**?**), though grid-like decoding can be artefactual without careful controls (**?**). This literature studies one condition at a time; we go comparative across the encoder-bandwidth spectrum.

Cognitive map theory and place-cell remapping The cognitive map hypothesis (**??**) casts internal models of space as flexible navigation substrates. Modern accounts model hippocampal codes as predictive maps over future state visits (**?**) and relational generalisers binding sensory observations to abstract structure (**??**). Place-cell rate maps shift along a remapping spectrum — no remapping (all fields shared between contexts), rate remapping (same fields, different rates), and global remapping (no shared fields) (**?**) — which **?** formalise as hidden-state inference (place fields remap when sensory evidence supports distinct contexts). Comparative cognition documents how sensor structure shapes navigation across species: blind subterranean rodents path-integrate without vision (**?**), echolocating bats build cognitive maps (**??**). We adopt this analysis vocabulary — Skaggs information (**?**), the remapping spectrum — and read the four sensor conditions as silicon analogs of distinct sensory niches.

We draw on three interpretability families: probing classifiers with selectivity controls (**??**); representational geometry (**????**); and causal/behavioural intervention on the recurrent state (memory transplant and shortcut discovery in this work; principled upgrades for follow-up: distributed alignment search (**?**) and Sussillo–Barak fixed-point analysis (**?**)). Convergence across the three deployed families on the same condition-level contrast is the strongest claim our methodology supports.

3 Methods

3.1 Experimental setup

Task and training. PointGoal navigation in Habitat (**?**) trained with DD-PPO (**?**). Episodes terminate on success (within 0.2 m of the goal) or 2000-step timeout; per-episode metrics use SPL (**?**). The training pool merges Gibson-0+ (**?**) (411 scenes) with Matterport3D-train (61 scenes); held-out evaluation uses 18 MP3D test scenes (1008 episodes) for distribution-shift probes. All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar). Uniform 250M-frame budget, single seed per condition — matching standard practice in the comparative-cognition modelling literature (**????**); rigour at the cross-condition level instead comes from convergent multi-method evidence (§**??**), supplemented by an independent-retraining replication of the blind condition (Appendix **??**). Full hyperparameters, action space, and architecture in Appendix **??**.

Visual sensor conditions. The four main conditions span an axis of encoder spatial bandwidth, listed from low to high:

- *Blind*: no visual stream (encoder bypassed).
- *Coarse*: 48×48 uniform RGB; ResNet-18's 32-fold spatial downsampling collapses the feature map to 1×1, removing per-step spatial structure while preserving channel information.
- *Foveated*: 256×256 RGB under eccentricity-dependent Gaussian blur ($\sigma_{\max}$=8, quadratic falloff, gaze fixed at image centre); 4×4 encoder feature map.
- *Uniform*: unblurred 256×256 RGB; 4×4 encoder feature map.

A separate *log-polar foveation control* (Appendix **??**) replaces Gaussian blur with log-polar resampling, yielding a $\sim 2\times2$ feature map; this isolates the encoder-spatial-output mechanism from blur per se and addresses the residual confound that coarse's 48×48 input does not fully separate 1×1 feature collapse from low input resolution.

Probing data collection. After training all weights are frozen and we collect per-step top-layer LSTM hidden states ($\mathbf{h}_2 \in \mathbb{R}^{512}$) from $n = 500$ Gibson held-out episodes under deterministic (argmax-action) rollouts.[2] The probing methodology then follows the three blocks of Figure **??** (§**??**–**??**); convergence across the three is the strongest claim the methodology supports. Hyperparameters and protocol specifics are in Appendix **??**.

3.2 Decodability: what does $\mathbf{h}_2$ encode about position?

Position information must be in $\mathbf{h}_2$ before format claims are meaningful, and two confounds drive multi-method redundancy. First, a single static readout cannot separate linear-readable structure from memorisation (linear R^2 inflates) or from non-linear codes the policy cannot use (MLP R^2 overestimates). Second, GPS enters the LSTM as a per-step L0 input, so static decodability does not by itself distinguish integrated memory from passthrough; ruling out passthrough requires temporal tests passthrough cannot produce.

Snapshot decodability: what is encoded right now? Linear Ridge probes with *Hewitt–Liang label-permutation selectivity* (**?**) (memorisation control); 2-layer MLP probes (non-linear ceiling); MINE (**?**) for $I(\mathbf{h}_2; \text{position})$ as a probe-form-independent anchor; per-unit *Skaggs spatial information* (**?**) with a circular-shuffle null for place-unit counts.

Memory dynamics: integrated code or per-step passthrough? *Step-binned probing* (separate Ridge per log-spaced step-in-episode bin) for within-episode time-variation; *lag-k retrospective probe* (decode GPS_{t-k} from $\mathbf{h}_t$) for memory persistence; *predictive-horizon probe* (decode $(x, y)_{t+k}$) testing the successor-representation prediction of **?** — positive future R^2 is impossible under passthrough since future GPS is not yet an input; *excursion-forgetting probe* (mid-episode random-action detour and recovery) as a perturbation test.

[2] $\mathbf{h}_2$ is the policy-readable slice; full (L0, L1, L2)$\times$(hidden, cell) probes are in Appendix **??**. Stochastic-action rollouts collapse to ~ 4 steps via STOP entropy and are not used (Appendix **??**).

The two together locate the position code on a magnitude axis (linear/MLP/MI), a population axis (Skaggs), and a temporal axis (lag/horizon/recovery), supplying H1 magnitude, the place-cell vocabulary, and the memory-vs-passthrough distinction.

3.3 Intrinsic structure: do conditions share a code?

The format claim — conditions allocate to non-interchangeable subspaces of $\mathbf{h}_2$ — has two independent sides: *within-condition* (is one condition's linear position axis scene-invariant or scene-conditional?) and *across-condition* (do different conditions share a basis?); both must be measured.

Within-condition: is the linear axis scene-invariant? *Leave-one-scene-out (LOSO)* cross-validation operationalised as a no-remap vs. global-remap test in the sense of **?**; cross-scene preferred-bin Spearman between rate maps as a per-unit complement.

Across-condition: do conditions share a basis? Cross-condition probe transfer (train Ridge on A, test on B); *Linear CKA* (**?**); 1-NN purity; *generalised Procrustes shape distance* θ_1 (**?**), a true metric satisfying the triangle inequality (unlike CKA); principal angles between top-K PCA bases plus position-direction cosines; per-condition complexity via participation ratio and TwoNN intrinsic dimension (**??**).

The two together produce the H2/H3 format claim from two independent angles (scene-invariance dichotomy + cross-condition geometric distance), jointly hard to fake by single-method artefact.

3.4 Causal intervention: does the policy rely on $\mathbf{h}_2$?

Probe-readability is necessary but not sufficient for behavioural relevance: a code can be linearly readable yet policy-irrelevant, or non-linearly encoded yet policy-relied. The two blocks above are correlational at the representation level; consumption requires causal tests outside any probe framework.

Representational causation: can foreign or stale memory be used? *Memory transplant*: mid-trajectory paste of donor $\mathbf{h}_2$ into recipient across the full 5×5 donor-recipient matrix; a self-transplant (donor from a different episode of the same condition) controls for replacement disturbance. *Memory-init transplant*: re-initialise from end-of-previous-episode memory vs. zero, isolating trajectory-bound vs. scene-generic memory.

Behavioural tests: does intact memory change behaviour? *Shortcut discovery* (Tolman 1948 in silico, paired same-scene-different-goal episodes): SPL drop under reset vs. persistent LSTM indexes how tightly the policy couples actions to integrated memory. *GPS-sensor mid-rollout ablation*: zero GPS and compass from step 0, measure end-of-episode success.[3]

The two together dissociate probe-readable from policy-relied capacity; the transplant matrix and the shortcut SPL drop together populate the consumption axis of the synthesis (Figure **??**).

3.5 Scope

The protocol answers the three questions above but is not calibrated to (i) separate non-linear from linear codes beyond the MLP-probe ceiling (Sussillo–Barak fixed-point analysis (**?**) is left for follow-up work); (ii) establish causal mechanism beyond input–output transplant; (iii) estimate per-condition seed variance directly — single seed per condition follows the comparative-cognition modelling convention (§**??**), with one independent-retraining replication for the blind condition.

4 Results

4.1 Comparable navigation skill across all four conditions

Table **??** collects the headline per-condition numbers. **All four agents learned; none failed.** Sighted conditions reach 99% success and blind reaches 93%; a Bug-style controller without learning achieves

[3] Zero-input ablations push the LSTM off its training distribution, so we use them only at the policy-output (success) level; an in-distribution alternative (replacing GPS/compass with training-marginal samples) is left for future work.

Table 1: Per-condition headline. All agents trained to a uniform 250M frames (§**??**); blind uses an independently retrained checkpoint (Appendix **??**). Probe R^2 columns: 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). The mid-episode story differs from the long-tail story: all four sighted conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200, Figure **??**b); the rich-encoder long-tail collapse produces the sub-chance no-cap numbers here. Bold marks (i) blind/coarse rows where GPS+Compass remain decodable (H1 phenomenon; one-sample t-test against zero, $p < 0.001$) and (ii) best task performance. The wide σ in rich-encoder rows reflects probe-at-chance behaviour: when no linear signal is present, individual CV folds vary based on noise, and the mean GPS R^2 is statistically indistinguishable from zero (foveated 0.18 ± 0.66, $t \approx 0.60$, $p \approx 0.58$; uniform -1.79 ± 2.99, $t \approx -1.34$, $p \approx 0.25$; one-sample t-test, df=4). The H1 claim is therefore that linear top-layer readability *collapses to chance* in rich-encoder conditions, not that it is reliably negative.

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	340M	0.59	0.93	**+0.94±0.03**	**+0.81±0.08**
Coarse	250M	0.853	0.989	**+0.582±0.099**	**+0.659±0.061**
Foveated	250M	**0.894**	**0.993**	+0.178±0.659	+0.503±0.142
Foveated-logpolar	250M	0.879	0.990	−0.029±0.759	+0.467±0.330
Uniform	250M	0.891	**0.993**	−1.785±2.993	−1.244±3.579

SPL 0.07 on the same scenes — an order of magnitude below every trained agent (Appendix **??**). The narrow success bracket across conditions rules out a skill-gap account for any cross-condition internal difference below.

The four agents are shown different visual inputs but reach near-identical task competence on identical task and training. *They must therefore have learned different internal representations of the same task.* The open question the rest of §**??** addresses is *what kind* of representation each condition learned: how the recurrent state encodes spatial position, how those encodings differ across conditions, and what this says about the encoder–memory race.

4.2 Magnitude axis (H1): linear position code shrinks with encoder bandwidth

The H1 claim, precisely Rich visual input *appears to reduce the linear readability of the GPS-derived world-frame position signal at the policy-readable LSTM layer*; phrased in within-episode terms, rich-vision agents do not maintain a temporally stable, linearly readable top-layer position signal across full episodes. We do *not* claim rich-vision agents fail to encode positional information altogether: 2-layer MLP probes recover position from the same top-layer state at $R^2 \in [0.51, 0.73]$ in rich-encoder conditions (Appendix **??**). The H1 contrast is specifically about whether a *linear* top-layer readout — the operation the DD-PPO policy head performs — can recover position, not about whether position is encoded anywhere in the hidden state. Throughout this section "GPS code" is shorthand for "linearly decodable top-layer GPS-derived position signal".

The bandwidth–allocation tradeoff is visible in a single plot (Figure **??**): as encoder spatial output dimension grows, the top-layer linear GPS R^2 falls monotonically across the four main conditions, and the log-polar control at $\sim 2{\times}2$ falls into the rich-encoder regime rather than the bottleneck regime — a falsification of the simplest spatial-output-dimensionality reading and motivation for the spatial-feature-variety reframing in Appendix **??**. The contrast is specifically about world-frame position: goal-distance (a sanity-check probe target with the same dimensionality as a position scalar) decodes at $R^2 \in [0.81, 0.90]$ in every condition, ruling out a probe-capacity reading.[4]

Information allocation: total info shifts gradually, linear-readable info shifts much faster Capacity allocation between encoder and memory routes makes a graded prediction about total position information in $\mathbf{h}_2$: it should decrease with encoder bandwidth (some pos-info shifts from $\mathbf{h}_2$ to encoder route), but the linear-readable component should decrease much faster (rich-encoder agents push the residual position code into non-linear $\mathbf{h}_2$ directions). Three measurements give this prediction quantitative shape (Figure **??**, Appendix **??**). Linear (Ridge $\alpha = 10$) GPS R^2 varies dramatically across the bandwidth axis ($+0.939$ blind, $+0.580$ coarse, $+0.162$ foveated, -0.121

[4] Whether the correlation reflects causation is what the §**??** experiments and the Appendix **??** resolution sweep test directly. All numerical claims in §**??** are single-seed per condition (cogsci modelling convention; see §**??**).

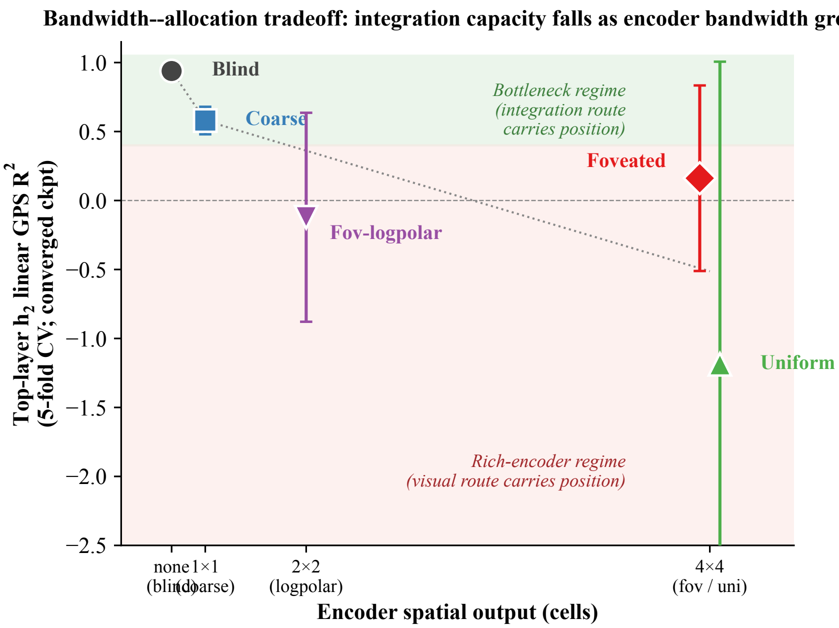

Figure 2: **The capacity-allocation principle, quantitative form.** Top-layer $\mathbf{h}_2$ linear GPS R^2 (5-fold CV mean $\pm\sigma$) plotted against encoder spatial output dimension. All sighted conditions at 250M frames; blind at 340M (Appendix **??**). Conditions: blind (no encoder), coarse (1×1), foveated-logpolar ($\sim 2\times2$, Appendix **??**), foveated (4×4 blurred), uniform (4×4). The monotonic anti-correlation is the bandwidth–allocation tradeoff: as encoder spatial bandwidth grows, less integration capacity remains in the recurrent state. Foveated and uniform sharing x but differing in y shows the design space is not 1-D — channel-level information is a second axis, exploited in §**??**.

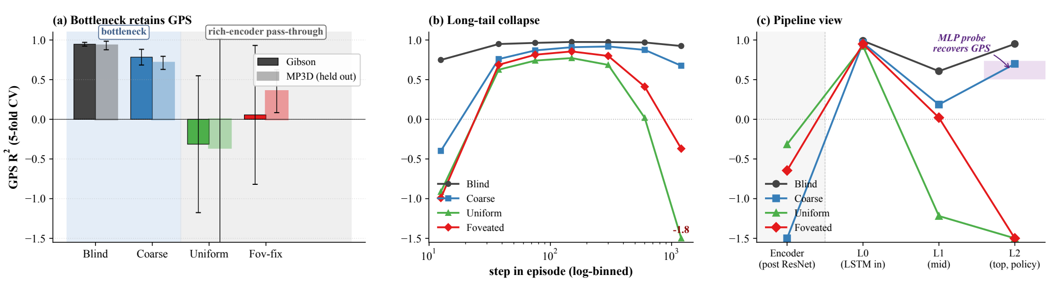

Figure 3: H1 evidence in detail. *(a)* Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; values outside y-range labelled). *(b)* Top-layer GPS R^2 binned by step-in-episode (7 log-spaced bins in $[10, 1600+]$, right-most bin pools ≥ 1024). *(c)* GPS R^2 along Encoder $\rightarrow$ L0 (sensor concat) $\rightarrow$ L1 $\rightarrow$ L2 (policy-readable). Shaded band at L2: 2-layer MLP probe recovery range ($R^2 \in [0.51, 0.73]$). Blind has no encoder.

foveated-logpolar, -1.187 uniform; range ≈ 2.1 across the five conditions). A 2-layer MLP probe (hidden $= 256$, $L_2 = 10^{-4}$, same 5-fold episode-level CV) compresses the range substantially ($+0.985$ blind, $+0.797$ coarse, $+0.763$ foveated, $+0.335$ foveated-logpolar, $+0.482$ uniform; range ≈ 0.65). Direct mutual-information estimation via MINE (**?**) (Appendix **??**) gives a nat-scale anchor on the same data. The gap between MLP and linear R^2 widens with encoder bandwidth: $+0.047$ blind, $+0.217$ coarse, $+0.602$ foveated, $+0.457$ foveated-logpolar, $+1.669$ uniform. The bottleneck end of the axis (blind, coarse) has near-zero gaps and the rich-encoder conditions much larger gaps — information present in rich-encoder $\mathbf{h}_2$ is mostly non-linear. This is the precise form capacity allocation takes: *total recoverable position information shifts gradually with encoder bandwidth ($\approx 1.5\times$ from blind to uniform via MINE; Appendix **??**), while linear-readable information shifts much faster (a ≈ 2.1-magnitude collapse in linear R^2). Bottleneck conditions concentrate*

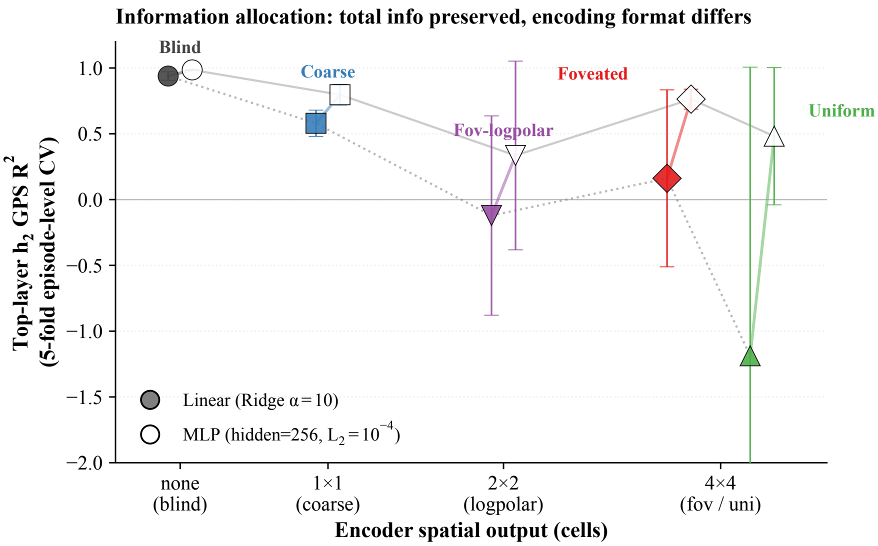

Figure 4: **Information allocation: linear vs. MLP readout.** Per-condition top-layer $\mathbf{h}_2$ GPS R^2 under linear (filled markers, dotted line) and 2-layer MLP probing (hollow markers, solid line); same episode-level 5-fold CV, 20 000 episode-step samples per condition. Linear R^2 range ≈ 2.0 across conditions; MLP range ≈ 0.5. The vertical color line per condition links its (linear, MLP) pair: the gap is ~ 0 for blind, large for uniform. Position information is approximately preserved by MLP probing in rich-encoder conditions even when linear readability collapses — a format shift, not a total-information collapse. MINE-estimated total $I(\mathbf{h}_2; \text{pos})$ shows a more modest $\approx 1.5\times$ decrease across conditions (Appendix **??**); we do not claim strict information conservation.

position into a linear top-layer direction; rich-encoder conditions push the residual position code into non-linear $\mathbf{h}_2$ *directions while also offloading some to the visual encoder route.*

Temporal stability is the precise claim To probe within-episode dynamics, we split each episode into seven log-spaced bins by step-in-episode (steps 10–50, 50–100, 100–200, ..., 1024+) and fit a separate Ridge probe per bin (*step-binned probing*, Figure **??**b). Rich-encoder conditions show a striking within-episode pattern: top-layer GPS R^2 is high (0.6–0.8) in the typical mid-episode window (steps 50–200) but drops near zero — statistically indistinguishable from chance, the level a probe achieves predicting the per-step mean — in the long-tail bins (steps 1024+). The headline "rich-encoder near chance" in Table **??** is therefore an average dragged down by the long-tail bins, not a mid-episode absence. *The H1 phenomenon is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code: rich-encoder agents establish a linear position code mid-episode but fail to maintain it.* A complementary *lag-k probe* (decoding the position k steps in the past, GPS_{t-k}, from the current hidden state $\mathbf{h}_t$) corroborates the bottleneck-vs-rich-encoder dichotomy: blind sustains $R^2 \geq 0.94$ across $k = 0$–20 ($+0.95, +0.95, +0.95, +0.94, +0.88$ at $k = \{0, 2, 5, 10, 20\}$); coarse is also stable but at lower magnitude ($+0.58, +0.58, +0.57, +0.56, +0.53$, std ≤ 0.18). Rich-encoder conditions show unstable lag-k profiles: foveated drifts from $+0.18$ at $k = 0$ through $+0.14$ at $k = 10$ to -0.07 at $k = 20$; uniform is strongly sub-baseline already at $k = 0$ ($R^2 = -1.78 \pm 2.99$) and degrades further with lag ($R^2 \approx -3.01$ at $k = 10$, -6.53 at $k = 20$); foveated-logpolar tracks foveated through the lag range (-0.03 at $k = 0$ to -0.12 at $k = 20$). The rich-encoder instability we attribute to trajectory-phase statistics rather than maintained memory (Appendix **??**).

Pipeline view: where GPS sits, and where it disappears. Tracing GPS R^2 from the encoder feature-map up to the policy-readable top layer (Figure **??**c) localises the H1 divergence. The encoder is at chance for all sighted conditions, so H1 is not an encoder-level retention property — encoders are roughly equally GPS-blind. The L0 jump is the GPS-sensor embedding entering the LSTM input vector, not the network computing position. The divergence is specifically at L2, the policy readout: bottleneck conditions maintain a high linear R^2 for the L0→L2 propagation, while rich-encoder conditions do not. Since the DD-PPO action head is a linear projection of $\mathbf{h}_2$, "linearly decodable at L2" maps directly to "policy-accessible". A 2-layer MLP on the same L2 states does recover GPS for rich-encoder conditions, so H1 is precisely about *linear* top-layer readability, not absence of GPS information.[5] The blind result replicates Wijmans et al. (**?**); coarse extends their finding to a condition with visual input but a 1×1 encoder output — the shared trigger is minimal visual-feature variety per step, not absence of vision.

Capacity-reallocation mechanism, with cross-training evidence The MLP recovery shows GPS information is present at L2 in rich-encoder conditions; the question is why the LSTM stops carrying it linearly. We propose H1 is a *capacity-reallocation* effect: integrating the L0 GPS-sensor input across time and exploiting the encoder's position-correlated visual features are two routes competing for finite hidden-state capacity. Bottleneck conditions force capacity to the integration route (no rich visual features available), and the temporally-stable GPS code persists in $\mathbf{h}_2$. Rich-encoder conditions provide a viable visual route; policy gradient reallocates capacity to it, and the integrated code is no longer linearly readable from the top layer — not absent, just spent elsewhere.

Cross-training probes at 5 training checkpoints per condition (Figure **??**) support this account. All four sighted conditions show clear early-positive GPS R^2 at ~ 50M frames (coarse $+0.91$, foveated $+0.91$, uniform $+0.84$, foveated-logpolar $+0.92$; tight 5-fold spread comparable across conditions), establishing that the linear top-layer code emerges early in training across the bandwidth spectrum. Bottleneck (coarse) preserves R^2 in $[0.43, 0.58]$ through training. The three rich-encoder conditions decay: uniform's R^2 leaves tight-positive by ~ 100M frames ($+0.79$ at ckpt-20), descends through 0 by ~ 200M (-0.39 at ckpt-40), and reaches catastrophic $R^2 \approx -1.79$ by 250M. Foveated decays slower, sitting at $R^2 \approx +0.53$ through $\sim$ 100–200M frames before declining to $+0.18$ at 250M. Foveated-logpolar (a third rich-encoder condition with 2×2 encoder spatial output) tracks foveated through training and lands at -0.03 at 250M — closer to uniform/foveated's rich-encoder regime than to coarse's bottleneck regime, despite its smaller encoder spatial output (Appendix **??**). Substitution timescale tracks encoder informativeness about position — uniform substitutes fastest, foveated and foveated-logpolar slower. The early-then-decay contrast is both within-episode (Figure **??**b) and across training: rich-encoder agents acquire a temporally-stable linear position code transiently, then that linear readability declines as the visual route consolidates. Compass codes follow the same bottleneck-vs-rich-encoder pattern (Figure **??**b), broadening the substitution claim from GPS specifically to spatial codes generally; uniform's compass linear R^2 remains positive ($\approx +0.36$ at convergence) while its GPS linear R^2 collapses to chance, suggesting heading is the more linearly-recoverable component of the visual-route substitute.[6]

Place-cell signature: similar per-unit Skaggs but different place-unit counts At the population level we apply the canonical Skaggs spatial-information formula (**?**), $I = \sum_b p(b)\,(\lambda_b/\bar{\lambda})\,\log_2(\lambda_b/\bar{\lambda})$, to each LSTM unit using rectified $\max(\mathbf{h}_2, 0)$ as the proxy firing rate λ_b (Figure **??**). The canonical metric is essentially condition-flat in the sighted regime: mean per-(unit, scene) Skaggs is 1.20 (coarse), 1.24 (foveated), 1.19 (uniform), 1.16 (foveated-logpolar) — a ≈ 0.08-bit range across the four sighted conditions; blind sits slightly higher at 1.38. Place-unit counts (> 1 bit) reverse the gradient: blind has *fewer* place units per scene (259) than the four sighted conditions (289 coarse / 303 foveated / 285 uniform / 284 foveated-logpolar). The population reading is that blind concentrates spatial information into fewer, more-informative units; sighted conditions distribute it across a broader population. We read the higher sighted place-unit count as view-cell-like binding to scene-specific visual context (**?**) rather than additional allocentric place coding (the LOSO scene-invariance result, Figure **??**, captures the latter and goes the other way). Per-unit Skaggs alone does not discriminate bottleneck from rich-encoder regimes: the bandwidth–allocation signal lives in linear-readout scene-invariance (Figure **??**) and information allocation (Figure **??**), not raw per-unit spatial bits.[7]

[5] The single-seed cross-dataset shift on MP3D — foveated GPS at chance $\to +0.35$ — awaits multi-seed replication; if it survives, it would suggest a condition-specific MP3D recovery worth investigating, but at $N = 1$ we cannot rule out per-seed noise. The MLP result alone cannot exclude exploitation of position-correlated features; Appendix **??** discusses both.

[6] Per-checkpoint probes are single-seed per condition (see §**??**); the decay-rate ordering is the most seed-exposed quantity in this section. A behavioural causal probe (mid-rollout GPS-sensor ablation) is reported in §**??**; a representational causal probe (mid-rollout encoder-output zeroing, which severs the visual route in-manifold) is left for future work.

[7] Place-vs-view-cell dissociation under cross-scene rate-map stability (**?**) would tighten this reading directly and is left for follow-up.

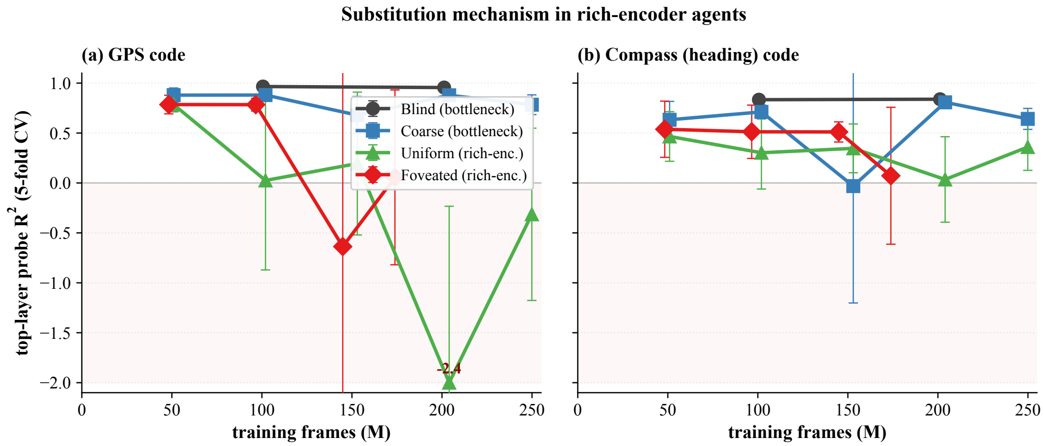

Figure 5: Cross-training probe profile across 4–5 training checkpoints per condition, truncated at 250M frames. *(a)* Top-layer GPS R^2. *(b)* Top-layer compass R^2. (500 ep/probe, 5-fold CV, single seed; hollow dotted = partial-data ckpts; negative-R^2 outliers clipped at -2.0.)

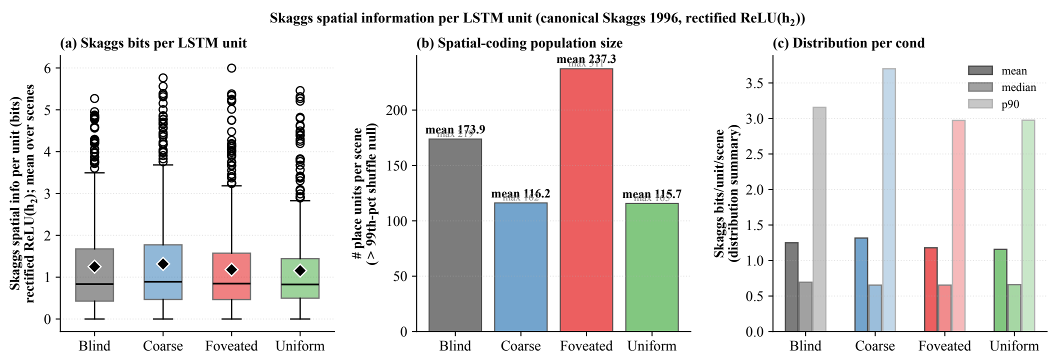

Figure 6: **Skaggs spatial information per LSTM unit.** *(a)* Distribution of per-unit Skaggs bits (averaged across 15 top scenes per condition); bottleneck and rich-encoder conditions overlap substantially (means ≈ 1.16–1.32 bits). *(b)* Number of place units per scene exceeding the 99th percentile of a 100-shuffle null; foveated has the largest count (≈ 237/scene), blind/coarse/uniform ≈ 116–174. (c) Distribution summary; cross-condition differences sit within the within-condition spread. The canonical Skaggs metric on rectified activations does not by itself discriminate bottleneck from rich-encoder regimes.

Predictive horizon: bottleneck conditions encode upcoming positions linearly Predictive-horizon probes from $\mathbf{h}_t$ to future positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$ test the successor-representation prediction of **??** that hippocampal codes encode discounted future occupancy (Appendix **??**). The signature is asymmetric across the bandwidth axis. Blind sustains a long predictive horizon: linear $R^2 \geq 0.94$ from $k = 0$ to $k = 20$ (only a 0.06-point drop) and $+0.79$ at $k = 50$. Coarse follows the same pattern at $R^2 \in [0.37, 0.58]$. Foveated decays from $+0.18$ at $k = 0$ to near zero by $k = 20$. Uniform and foveated-logpolar have no predictive horizon: linear R^2 is already negative at $k = 0$ and grows more negative; MLP probes recover at most ~ 0.5. Bottleneck conditions encode $\mathbf{h}_t$ as a smoothly time-varying integrated code that linearly predicts upcoming positions (the SR-style cognitive-map signature); rich-encoder conditions encode $\mathbf{h}_t$ reactively to current visual context without temporal extrapolation.

4.3 Format axis (H3): scene-invariance distinguishes bottleneck from rich encoders

The linear→MLP gap (Figure **??**) shows position information is approximately preserved by MLP probing across conditions even when linear readability collapses — a format shift rather than

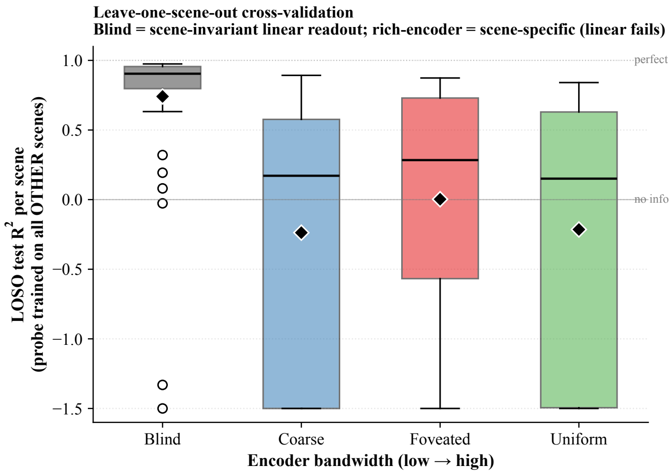

Figure 7: **Leave-one-scene-out (LOSO) cross-validation.** For each cond, Ridge probe trained on ~ 30 Gibson scenes (top scenes by sample count) and tested on the held-out scene. Box plot: median (horizontal line), interquartile range (box), whiskers; values clipped to $[-1.5, 1]$ for visualisation. Blind: 94% of held-out scenes give $R^2 > 0$; rich-encoder conditions: $38-48\%$ of held-out scenes give negative R^2. Encoder bandwidth dictates whether the linear position-axis is scene-invariant (linearly probable across scenes) or scene-conditional (requires non-linear readout to switch contexts).

a total-information collapse (MINE shows a more modest $\approx 1.5\times$ decrease in total $I(\mathbf{h}_2; \text{pos})$; Appendix **??**). What changes *format* between bottleneck and rich-encoder conditions? A leave-one-scene-out (LOSO) cross-validation localises the answer: for each held-out Gibson scene, fit Ridge on the other ~ 30 scenes and report test R^2 on the held-out scene (Figure **??**). Blind achieves median LOSO $R^2 = 0.92$ with 0% of held-out scenes giving negative R^2; coarse drops to median 0.17 (48% scenes negative); foveated 0.28 (38%); uniform 0.15 (44%). The bottleneck-vs-rich-encoder rank ordering is preserved across the cross-training trajectory (Figure **??**), confirming the LOSO contrast is not specific to the 250M endpoint.

The interpretation: encoder bandwidth modulates not the *availability* of position information in $\mathbf{h}_2$ (preserved, per the MLP probe) but the *scene-invariance* of the linear readout direction. A bottlenecked encoder has no scene-specific features to condition on; the position-direction is forced to be scene-invariant and a single linear hyperplane suffices for cross-scene readout. A rich encoder lets the LSTM use scene-specific visual features to encode position; different scenes induce different position-directions in $\mathbf{h}_2$, and a single linear probe cannot capture this scene-conditional encoding while an MLP can by switching its readout on visual context. This re-states capacity allocation in IB-invariance terms (**?**): bottleneck-induced encoding is invariance-disentangled, rich-encoder encoding is scene-conditional disentanglement.

4.4 Format axis (H2): conditions allocate to non-interchangeable subspaces

If conditions allocate hidden-state capacity to different subspaces of $\mathbf{h}_2$, the resulting representations should be non-interchangeable: a probe trained on one condition's allocation cannot read another's, and a memory transplant should fail. Two complementary tests confirm this.

Behavioural test (memory transplant) Pasting a donor's mid-episode hidden state into a recipient's policy at step 30 tests whether the representational geometry differences *matter* outside the probe framework (Figure **??**, right). Three structural facts emerge that no linear-similarity measure on its own would surface.

(i) Direction-of-transplant asymmetry. The transplant matrix is markedly asymmetric. When a *bottleneck* condition (blind / coarse) provides the donor state and a *rich-encoder* condition (foveated / uniform) is the recipient, recipient SPL collapses by 0.3–0.5. Run in the reverse direction — rich-encoder donor, bottleneck recipient — recipient SPL is largely preserved. In plain terms: *rich-encoder*

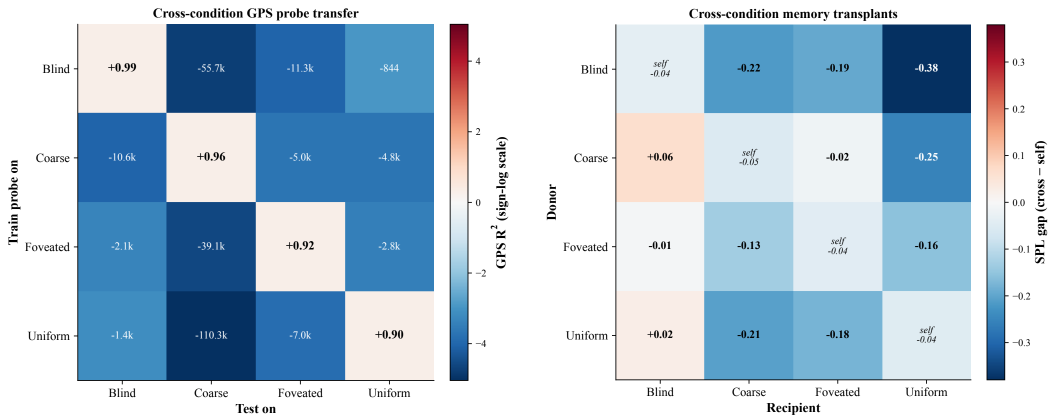

Figure 8: Subspace divergence (H2), two views. *Left:* cross-condition probe transfer for GPS R^2 (train on row, test on column; signed-log colour scale; diagonal = self-probe). *Right:* $4{\times}4$ cross-condition memory-transplant matrix at midpoint $t{=}30$; off-diagonal = cross-transplant SPL − recipient's self-transplant SPL. Single seed; midpoint-sweep companion in Appendix **??**.

policies cannot make use of bottleneck-style hidden states, but bottleneck policies can still drive themselves from rich-encoder hidden states.

(ii) Recipient sensitivity scales with encoder bandwidth. Uniform-recipient SPL drops the most under *any* foreign donor; blind-recipient SPL drops the least. The richer the encoder, the more brittle the policy is to memory replacement.

(iii) Linear similarity misses these effects. Condition pairs that read as indistinguishable under unaligned linear CKA still differ by 0.01–0.38 SPL under transplant. "Looks the same under linear-similarity" does not imply "has the same policy-relevant content".[8]

Convergent evidence at the probe level. Cross-condition probe transfer (Figure **??**, left) shows the same asymmetric pattern: probes trained on bottleneck states predict off-manifold by a wider margin on rich-encoder hidden states than the reverse. The 1-NN purity is 1.000 vs. chance 0.25 on $1500{\times}4$ pooled top-layer states — a sample's nearest neighbour by Euclidean distance is always within its own condition. The same 1.000 holds at $10\,000{\times}4$ pooled samples (a $6.7{\times}$ enlargement), ruling out a sample-size artefact. Together with the symmetrically catastrophic transfer failure (every off-diagonal cell), this rules out both the "one code embedded in another" alternative and the "same code at different scale" alternative — scaled embeddings would still admit cross-probe transfer. *Probe overfit does not account for the pattern*: all probes use 5-fold episode-level CV (folds are disjoint sets of episodes, so there is no within-episode train/test leakage); self-probes succeed at $R^2 \in [0.6, 0.95]$ within each condition (showing the probe family has the capacity to fit the hidden states); and Hewitt–Liang label-permutation selectivity (§**??**) tracks each probe's R^2 (the probe is reading real structure, not memorising arbitrary labels).[9]

Geometric evidence: shape-metric distances and intrinsic-dimension scalars Linear CKA tells us conditions look "not similar" but provides neither a true metric nor a complexity scalar; we add two methodologically distinct geometric views (Table **??**). *Shape-metric distance matrix* (**?**): the Procrustes Riemannian angular distance θ_1 on Kendall's shape space (computed on 500 paired episode-mean $\mathbf{h}_2$ vectors per condition, optimal-rotation-aligned in closed form). θ_1 is a *true metric* — it satisfies the triangle inequality and embeds in Euclidean space, two properties linear CKA cannot guarantee — and the pattern matches the bandwidth-allocation prediction: the bottleneck pair (blind–coarse, $\theta_1 = 1.10$) and the rich-encoder pair (foveated–uniform, $\theta_1 = 1.08$) are each tighter than cross-regime pairs (blind–foveated 1.23, blind–uniform 1.22, coarse–foveated 1.14, coarse–uniform 1.13). *Within-condition complexity scalars* (**?**): participation ratio (PR) and TwoNN intrinsic

[8] Single-seed; the asymmetry pattern is qualitatively robust across the full $4{\times}4$ matrix but specific cell magnitudes will move with replication.

[9] Boundaries: similarity tests are linear or near-linear, so non-linear measures could reveal higher-order shared structure; the four sensors are diverse but not exhaustive. Task competence is bunched (96–99%, §**??**), so the divergence is not a skill difference.

Table 2: **Representational geometry: within-condition complexity and cross-condition shape distance.** Each row is a condition. *Within-condition complexity (left three columns).* PR = participation ratio, the effective number of dimensions used by the representation (larger = variance spread across more directions); ID-MLE and ID-CDF are two TwoNN estimators of the manifold's intrinsic dimensionality (the two agree within ≈ 0.15). Bold marks the column-wise minimum: *coarse* separates from the other three on *both* measures, consistent with its $1{\times}1$ encoder feature map collapsing representational dimensionality. *Cross-condition shape distance (right four columns).* Procrustes Riemannian angular distance θ_1 between each pair of conditions' top-PCA shapes (range $[0, \pi/2] \approx [0, 1.57]$; larger = more distinct). The pattern is bandwidth-allocation-aligned: the rich-encoder pair *foveated–uniform* is closest ($\theta_1 = 1.08$), the bottleneck pair *blind–coarse* is next ($\theta_1 = 1.10$), and cross-regime pairs (blind–foveated, blind–uniform) are most distant ($\theta_1 \in [1.22, 1.23]$). Single seed; 500 paired Gibson eval episodes.

	Within condition			*Procrustes θ_1 to other condition*			
Condition	PR	ID-MLE	ID-CDF	Bl	Co	Fv	Un
Blind	4.10	1.99	1.85	–	1.10	1.23	1.22
Coarse	**3.10**	**1.76**	**1.64**	1.10	–	1.14	1.13
Foveated	**4.34**	1.89	1.77	1.23	1.14	–	1.08
Uniform	4.01	1.91	1.80	1.22	1.13	1.08	–

dimension. Coarse separates from the other three on *both* (PR 3.10 vs. 4.01–4.34; TwoNN-ID 1.64 vs. 1.77–1.85, CDF estimator), supporting that the $1{\times}1$ encoder collapse is structurally pulling down the LSTM's representational complexity, not just its decodability. Foveated has the highest PR (4.34); blind (4.10) sits between the rich-encoder pair and coarse, consistent with blind using extra dimensions to carry the integrated GPS code that other conditions either substitute (rich-encoder) or compress (coarse).[10]

Direct subspace test: principal angles and position-direction alignment The capacity-allocation principle predicts that conditions occupy mutually orthogonal subspaces of $\mathbf{h}_2$, with position encoded along non-aligned directions. We test this directly (Figure **??**). For each condition, we extract the top-K PCA basis of $\mathbf{h}_2$ ($K \approx 8$–10, covering $\sim 90\%$ variance) and the position-encoding direction (Ridge probe weight $\boldsymbol{\beta}$ for GPS). Across all six condition pairs, mean principal angles between top-K subspaces lie in $[86, 89]$ — maximally close to orthogonal in this hidden-state geometry — and pairwise cosines between position-encoding directions lie in $[-0.04, +0.02]$, indistinguishable from random. Conditions allocate to mutually orthogonal subspaces, and within those subspaces position is encoded along non-aligned directions. This is the subspace-divergence prediction in its sharpest form.[11]

Capacity allocation is established early, refined late The convergence-time orthogonality in Figure **??** could in principle reflect either (i) a late-emerging differentiation that builds slowly through training or (ii) a structural commitment laid down early and refined toward maximal orthogonality. Probing intermediate checkpoints distinguishes the two (Figure **??**). Across all six condition pairs and four to five training stages spanning ~ 50M to ~ 350M frames, we find: (a) principal angles between top-K PCA subspaces are already in $[83, 89]$ at the earliest probed checkpoint (~ 50M frames) and refine modestly to $[88, 90]$ by convergence (4–7° increase per pair); (b) position-encoding direction cosines are essentially uncorrelated from the earliest probed checkpoint (range $[-0.10, +0.06]$ throughout, indistinguishable from chance). The capacity allocation that produces non-interchangeable subspaces is therefore a training-time process whose direction is committed early; the convergence-time orthogonality measured in Figure **??** is a refinement of structure already present at ~ 50M frames, not a late-emerging novelty.[12]

[10] Single seed per condition (cogsci modelling convention; see §**??**). Procrustes pairing is per-episode mean (alternative pairings tabulated in Appendix **??**); ID estimators are MLE and Facco-CDF, agreeing within 0.15.

[11] Single seed per condition (cogsci modelling convention; see §**??**). Random pairs in 512-d would have expected principal angles slightly below 90° due to finite K; our values match this null up to 1–4°, indicating the subspaces are statistically indistinguishable from random orientations.

[12] Single seed per condition (cogsci modelling convention; see §**??**). Earliest probed checkpoint per condition is ckpt 5–10 ($\sim$ 50M sighted, $\sim$ 50M blind); whether sub-50M training shows further divergence is unprobed.

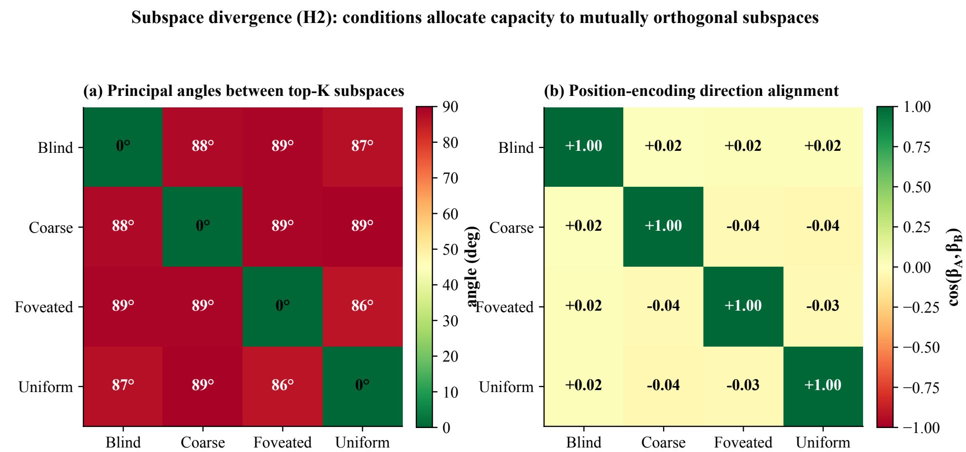

Figure 9: **Subspace divergence (H2), quantitative form.** *(a)* Mean principal angles between top-K PCA subspaces of $\mathbf{h}_2$ ($K \in \{8, 9, 10\}$ per condition, covering $\sim 90\%$ variance; $20\,000$ episode-step samples per condition). All off-diagonal angles in $[86, 89]$ — conditions occupy near-orthogonal subspaces. *(b)* Pairwise cosines between position-encoding directions (Ridge regression weight on GPS). All off-diagonal cosines in $[-0.04, +0.02]$ — position is encoded along essentially uncorrelated directions across conditions. Together, these quantities give the capacity-allocation prediction in direct form: conditions allocate hidden-state capacity to mutually unreadable subspaces, and the within-subspace position direction differs as well. Single seed.

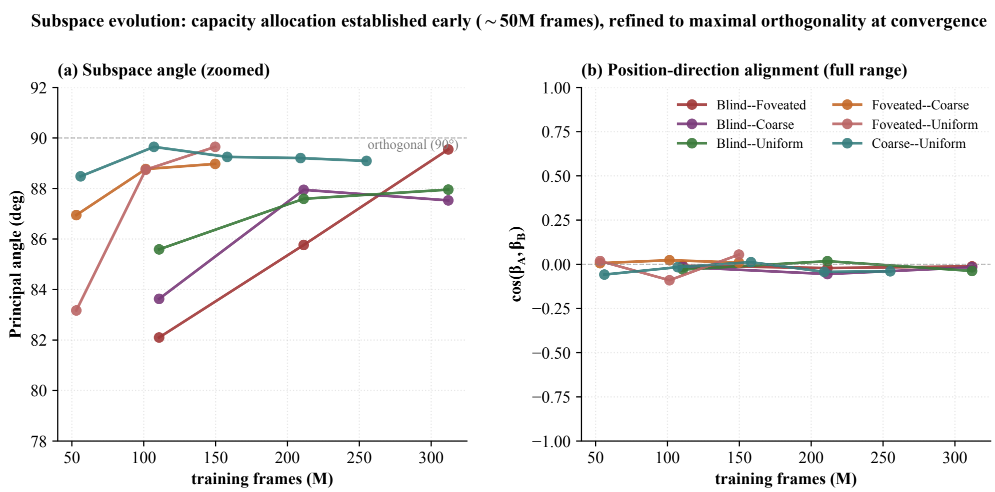

Figure 10: **Subspace evolution over training.** *(a)* Mean principal angle between condition-pair top-K subspaces, plotted across training checkpoints (~ 50M to ~ 350M frames; pairs matched at corresponding ckpt indices). Angles climb modestly from ~ 83–87 to ~ 88–90 over training, refining toward maximal orthogonality. *(b)* Position-encoding direction cosines are ~ 0 throughout, indicating direction divergence is committed before the earliest probed checkpoint. Together: capacity allocation is established early (~ 50M frames) and refined late, not a late-emerging divergence. Single seed; 4–5 ckpts per condition.

4.5 Foveation as a hybrid case: shared magnitude with uniform, distinct geometry

Foveation's allocation is hybrid: it groups with uniform on the bandwidth–allocation tradeoff (rich-encoder magnitude, §**??**), but separates from uniform on every other measure (Table **??**) — consistent with the capacity-allocation principle that the visual and integration routes are independent sinks rather than two ends of one knob, so foveation can spend capacity differently from uniform within the rich-encoder regime.

Table 3: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur, same ResNet-18 encoder. Sources: Table **??**, Figures **??**, **??**, **??**c. Single seed, deterministic rollouts.

Measure	Foveated	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, *indistinguishable*
Encoder $\to$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	**differ by 0.71**
Shortcut SPL drop %	21%	41%	**uniform 2× more brittle**
Memory-transplant cost (fov $\to$ uni)	-0.10 SPL beyond self-noise		**not interchangeable**
Pairwise CKA(fov, uni)	$< 10^{-4}$		**near-orthogonal subspaces**
1-NN purity (pooled)	1.000 vs. chance 0.20		**linearly separable**

Whatever foveation does differently from uniform happens inside the LSTM, in subspaces a linear top-layer GPS probe is blind to. The most informative single contrast is dataset shift: foveated agents recover a held-out-MP3D GPS code that uniform agents do not (a 0.71-magnitude gap, Table **??** row 4) — a *transferability* difference, not just two different codes on the same training scenes. Behavioural shortcut reliance, cross-condition transplant cost, and CKA / 1-NN purity all read foveated $\neq$ uniform in the same direction. The log-polar foveation control (Appendix **??**) was the falsifiable next step: the preregistered prediction was that a $\sim 2\times2$ encoder spatial output would land in the bottleneck regime ($R^2 \geq 0.3$) if the simple "encoder-spatial-output dimensionality" reading of H1 were correct. The result lands at $R^2 = -0.029$ — in the rich-encoder regime — falsifying that simple reading; the corrected mechanism is encoder spatial-feature *variety* per step (Appendix **??**).[13]

4.6 Consumption axis: probe-readability dissociates from policy reliance

Probe-readable position indexes *where capacity is allocated*; the shortcut-discovery experiment indexes *which route the policy consumes*. Under the capacity-allocation principle, these are independent quantities and need not coincide. Paired same-scene-different-goal episodes (20 scenes × 10 pairs) compare second-episode SPL under reset vs. persistent LSTM; the SPL drop indexes how tightly the policy couples actions to integrated recurrent memory. Plotted against probe-readable top-layer GPS, two off-diagonal cases emerge. *Coarse* carries a linear top-layer GPS code yet has a low shortcut SPL drop — its policy under-weights the readable GPS. *Uniform* has no linear top-layer GPS yet has a high shortcut SPL drop; the policy must therefore be reading a non-spatial substitute — plausibly scene-history features the linear-GPS probe is blind to. "Probe-readable" and "policy-relied-on" can therefore dissociate in either direction (visualised as the marker-size axis of Figure **??**, §**??**). We label these as candidate dissociations rather than established anomalies.[14]

Memory-init transplant test The §**??** memory-transplant experiment swaps a donor's hidden state into a recipient mid-episode. Here we test the converse: re-initialise the *same* agent at episode start with its own previous-episode-end memory $(\mathbf{h}_T, \mathbf{c}_T)$, and ask whether navigating S→T is helped or hurt by carrying scene-context from the trajectory just completed. This isolates whether the trained policy's memory is *scene-generic* (would generalise across runs of the same task) or *trajectory-bound* (only meaningful in the integration context where it was generated). All four conditions show *consistent negative* ΔSPL (range -0.10 to -0.14 across conditions; probe initialised with own final memory *vs.* zero-init agent baseline; single seed, n=150 episodes/condition). The robust direction across H1-divergent conditions says *the policy at* t=0 *expects a zero hidden state*; non-zero initialisation hurts navigation regardless of encoder type. This complements the subspace-divergence finding (§**??**): the capacity allocation is not just *cross-condition* non-interchangeable, it is *within-condition* trajectory-bound — the trained policy's mapping from memory to action is calibrated to the specific integration trajectory the LSTM rolled out, not to a scene-generic representation. Blind shows the largest absolute drop (success $0.81 \to 0.67$); the other three retain $\geq 90\%$ success but lose path-efficiency.[15]

[13] The foveation-vs-uniform contrasts in Table **??** are single-seed per condition (see §**??**).

[14] The 5 points are single-seed per condition (see §**??**); the dissociation pattern is supported by the convergent evidence in §**??**–**??**, but a full multi-seed grid would tighten the seed-variance bounds on individual off-diagonal points.

[15] Single seed per condition (cogsci modelling convention; see §**??**).

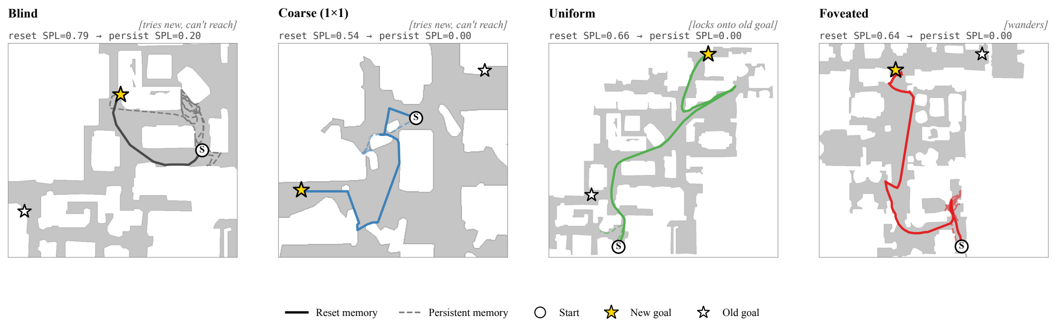

Figure 11: Persistent-memory failure modes — one canonical paired-episode case per condition (extended catalog: Appendix **??**). Aggregate per-condition margin = avg dist(persistent end $\to$ old goal) $-$ avg dist(persistent end $\to$ new goal) over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5$m): Blind -0.38m (n=27), Coarse -0.57m (n=35), Foveated -0.59m (n=16), **Uniform** $+1.83$**m (**n=46**, closer to OLD goal)**. Single seed.

If uniform's memory is non-spatial, what does it anchor to? Trajectory-level analysis of the persistent-memory failures gives a candidate answer (Figure **??**). Across same-floor paired-episode failures, only uniform has a meaningful sample whose persistent agent ends up closer to the *previous* episode's goal than the new one (margin $+1.83$m on n=46); blind, coarse and foveated all end nearer the new goal but fail to reach it.[16] This suggests uniform's persistent failure mechanism is a visual / scene-history anchor: the LSTM tells the policy "head to where you were going last episode". Blind's small negative margin (-0.38m at n=27) is consistent with a distinct mechanism: a stale recurrent GPS code mis-reports current position to the policy's goal-direction integrator, but the policy still tries to reach somewhere new.

Behavioural validation under GPS-sensor ablation A complementary behavioural test asks whether the policy reads the GPS sensor specifically (and which conditions tolerate its loss). We zero the GPS and compass inputs from step 0 in each rollout (the L0 sensor signals that drive the integration route), then measure end-of-episode success rate (n=50–100 episodes/cond, single seed). All four conditions collapse to 3–8% success (vs. baseline 93–99%), indicating that none of them navigates from purely visual landmarks — the GPS sensor is critical across the spectrum. The size of the drop modestly tracks the dissociation direction: *coarse* (highest probe-readable GPS, $R^2 = 0.78$) drops the most ($\Delta = -0.95$, success $0.98 \to 0.03$), and *uniform* (no probe-readable GPS, policy reading non-spatial substitutes) drops the least ($\Delta = -0.91$, success $0.99 \to 0.08$); foveated and blind fall between ($\Delta = -0.92$ and -0.87 respectively). The 4–8 pp inter-condition spread is small relative to the absolute drop and within plausible single-seed noise; we report this as consistent-but-not-conclusive support for the §**??** dissociation rather than independent confirmation.[17]

4.7 Boundaries: null results that bracket the principle

What does *not* discriminate also defines what our H1/H2 claims are about. Three null results bracket our findings. *(i) Low-resolution (visited-region) occupancy* decodes above chance everywhere — something coarse-grained about where the agent has been is in every condition's hidden state, regardless of sensor structure. *(ii) Metric* 20×20 *occupancy* fails everywhere (consistent with (**?**): high-resolution metric layout needs non-linear probes). *(iii) The ego-frame goal vector* is at chance everywhere, because PointGoal supplies it as a per-step input — the policy has no incentive to re-encode it. Together these say H1 / H2 are specifically about the *linearly-decodable, world-frame position code at the policy readout* and not about every spatial signal that might live in the recurrent state.

[16] Single seed per condition (cogsci modelling convention; see §**??**).

[17] Single seed per condition (cogsci modelling convention; see §**??**). The mid-rollout zeroed visual ablation (a stronger causal test of the bandwidth–allocation tradeoff in §**??**) was contaminated by the LSTM going off its training manifold under unseen zero-input distribution; we report only the behavioural success rate from GPS ablation, which is a policy-level signal independent of probe extrapolation.

Allocentric occupancy decoder (left for follow-up). A direct test of the "memory contains a metric map" reading would train a per-condition MLP decoder $f(\mathbf{h}_T, \mathbf{c}_T) \to$ binary occupancy grid (5-fold episode-level CV) and compare per-condition IoU. Ground-truth occupancy maps are already collected (106 scenes, `results/scene_occupancy/`); decoder train+eval is left for follow-up work and is the most direct extension of the present paper's results.

A counter-intuitive boundary observation comes from population coding (Appendix **??**): rich-encoder conditions have a few units with *higher* peak spatial information than blind, yet their hidden states do not linearly decode GPS. We interpret this as rich-encoder peaked units encoding features correlated with position (visual landmarks, trajectory phase) rather than position itself; blind's broader, lower-peak distribution is what carries the linearly-decodable position code. This is the inverse of the naive expectation that more peaked spatial tuning $\to$ better position decoding, and sharpens what H1 measures: *linear, top-layer, policy-readable* position, not per-unit spatial information.

Excursion-forgetting separates blind from the integrated-code regime A complementary perturbation test distinguishes *blind's* robustness from the *integrated-code* regime that coarse and the rich-encoder conditions share. We force a 25-step random-action detour mid-episode (50-step warmup $\to$ 25-step detour $\to$ 100-step recovery; $n = 100$ episodes/condition), train one Ridge probe on full-episode pooled hidden states (episode-level 5-fold CV), and measure scale-invariant prediction error (MAE / position-spread) per segment. All four conditions show *forgetting* (recovery error $>$ warmup error), but with markedly different magnitudes: blind $+0.17$, uniform $+0.31$, foveated $+0.34$, coarse $+0.43$ (Figure **??**). Blind's forgetting is roughly half that of the other three, consistent with its hidden state being closer to a per-step "GPS-sensor passthrough" (the sensor still reads correctly after the detour) than to an integrated path-integration code. The other three (including coarse, the other bottleneck condition) cluster around a more fragile integrated regime: coarse integrates GPS over time, rich-encoder agents commit some role to visual context — both decohere when the action stream is interrupted.[18]

5 Discussion

5.1 Synthesis: five views of one capacity allocation

The five predictions cohere as views of one capacity allocation (Figure **??**, summarised on three axes for visual compactness). The *magnitude axis (H1, X)* is how much capacity the integration route receives: bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones reallocate to the visual route (§**??**). The information-allocation, scene-invariance, predictive-horizon, and place-cell findings sharpen what this axis *means*: total recoverable position information shifts gradually with encoder bandwidth ($\approx 1.5\times$ from blind to uniform per MINE; Appendix **??**); what differs much faster is whether the position-axis is scene-invariant and linearly accessible (bottleneck) or scene-conditional and only nonlinearly accessible (rich encoder, Figure **??**); whether $\mathbf{h}_t$ encodes future positions linearly (bottleneck: yes, up to $k = 20$ steps; rich-encoder: no; Appendix **??**); and at the population level, per-unit Skaggs spatial information is similar across conditions but place-unit *count* and place-vs-view-cell character differ (Figure **??**). The *format axis (H2, Y)* captures which subspace the allocation lands in: cross-condition memory transplants are asymmetric, per-condition probes are mutually off-manifold, and probe-transfer R^2 is catastrophically negative across conditions — subspaces are not just orthogonal but jointly incompatible at the readout level (§**??**, Appendix **??**). The *consumption axis (marker size)* is the §**??** shortcut SPL drop: which route the policy actually reads. Each condition occupies a distinct (X, Y, size) coordinate, and the bandwidth–allocation tradeoff (H1) is one of three orthogonal axes, not the whole picture.

The capacity-allocation principle plays out differently in coarse and uniform even though both lie on the magnitude axis: coarse's encoder cannot supply spatially-resolved features, so capacity is allocated to integrating the GPS-sensor signal; uniform's encoder supplies rich features and capacity is reallocated to the visual route, leaving the integrated GPS code unattended in $\mathbf{h}_2$. The two are not the same mechanism, only two regimes of one allocation.

[18] Single seed per condition (cogsci modelling convention; see §**??**). The variance-matched MAE/spread metric we use corrects a prior R^2-based artefact where target-spread differences across segments inflated apparent recovery.

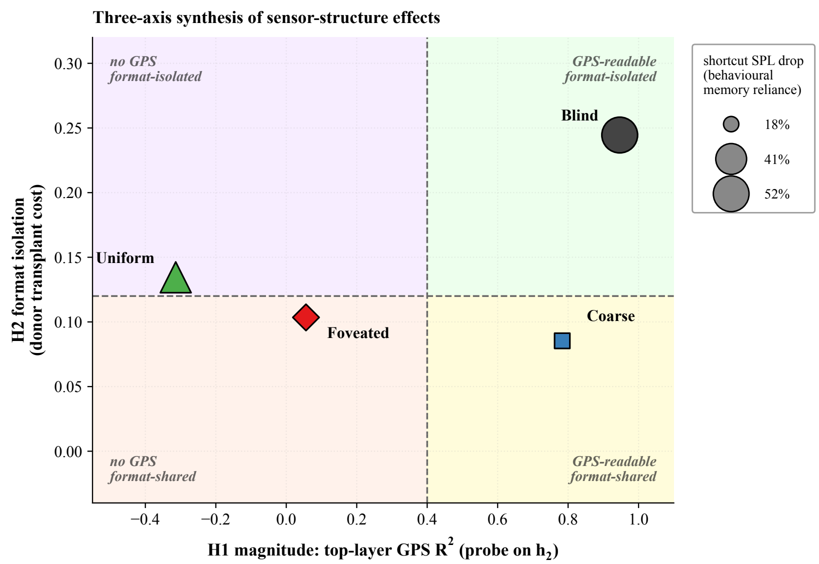

Figure 12: Synthesis: three orthogonal measurements on the same four conditions, chosen for the synthesis figure to summarise the high-level story. *X-axis:* top-layer GPS R^2 on $\mathbf{h}_2$ (H1 magnitude). *Y-axis:* $-1\times$ average cross-condition transplant cost when this condition is the donor (H2 format isolation; higher Y = more disruptive to other conditions' policies). *Marker size:* second-episode shortcut SPL drop $\%$ (larger = more policy dependence on persistent recurrent memory). The information-allocation, scene-invariance, and place-cell findings (Figures **??**, **??**, **??**) sharpen the *interpretation* of each axis but are not separate axes of variation.

5.2 Biological precedent: sensor structure shapes hippocampal representation

The hypothesis that sensor structure shapes spatial-memory format has long-standing precedent in comparative cognition, and individual biological observations consistent with it have accumulated over decades. We do not claim our silicon results *demonstrate* the bio-AI parallel; we claim only that the patterns we observe in our controlled setting — where the encoder, memory, policy, training distribution and reward are held fixed and only the visual sensor varies — are at high level consistent with what one would expect if the same hypothesis held in animals. Two threads of animal-navigation research are most closely related to our H1 and H2 findings; the dissociation finding (probe-readable vs. policy-used memory) does not have a directly established biological correspondence but echoes the broader theme that "having a representation" and "using it for behaviour" are dissociable in animal cognition.

H1 (sensor bottleneck $\rightarrow$ memory takes over). Congenitally blind humans show larger anterior hippocampal volume and supranormal performance on non-visual spatial-navigation tasks (**?**), alongside broader cross-modal recruitment of visually deprived cortex for non-visual processing (**?**); both are consistent with the hippocampus carrying more navigational load when the visual stream is absent. Rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates (**?**), indicating spatial behaviour can be sustained when the visual landmark stream is removed. These observations are at high level consistent with our blind / coarse regime, where the LSTM — deprived of rich visual variety per step — retains a linear top-layer GPS code that is no longer linearly readable in the rich-encoder conditions; we do not claim a direct bio–AI mechanistic correspondence.

H2 (different sensors $\rightarrow$ different spatial codes). The primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — has been attributed in part to the foveated-with-saccades vs. near-panoramic visual sampling regime each species uses (**?**). Cross-modality remapping in echolocating bats, where the same neurons recode under vision vs. echolocation (**?**), demonstrates that sensor modality is associated with shifts in the linear subspace of hippocampal coding. Both observations are at high level consistent with our H2 finding: the same task with different visual structure yields linearly separable LSTM subspaces (CKA $< 10^{-4}$, probe-transfer collapse, asymmetric memory transplant).

We treat these as convergent (not conclusive) evidence. Convergence across systems — and across silicon vs. neural — is suggestive of a sensor–memory coupling principle more general than any single architecture, but it does not show that the mechanisms are the same. Our experiments isolate the principle in a controlled, single-architecture setting; biological work establishes that the principle exists in animals at all. The two are complementary; neither is decisive alone.

Sensory-niche framing across taxa Read alongside the comparative-cognition literature, our four conditions can be loosely mapped onto canonical sensory niches without claiming a tight bio–AI correspondence. *Blind* is loosely modelled on subterranean rodents (blind mole-rat *Spalax*, naked mole-rat) that navigate via path integration plus magneto-reception in the absence of vision (**?**). *Coarse* (1×1 encoder) is closest to single-channel sensory summaries — echolocation acoustic features, infrared pit-viper-style scalar detectors — where each timestep contributes a low-spatial-bandwidth scalar but integration across time supports navigation. *Foveated* (4×4 with eccentricity-dependent blur) abstracts the primate / raptor regime of high-resolution central vision plus blurred periphery (we hold gaze fixed; gaze placement is a content axis of foveation we leave as future work). *Uniform* (4×4 unblurred) is closer to the insect compound eye / zebrafish near-isotropic visual sampling. The deep-RL framework provides a controlled experimental setting where these comparative-cognition observations (**??**) can be tested under within-architecture variation that animal experiments cannot directly run; the sensory-niche mappings above are interpretive scaffolding rather than claims of mechanistic identity. Cast in the hidden-state-inference framework of **?**, our LOSO result maps onto the no-remapping vs. global-remapping axis: the bottleneck conditions exhibit no remapping (a single scene-invariant linear position direction is shared across rooms; analogous to all place fields being shared between contexts), while the rich-encoder conditions resemble global remapping (no decoding direction is shared across rooms; analogous to no place fields shared between contexts). Under Sanders et al.'s account this matches the predicted role of sensory evidence in animals: limited visual variety per step does not support inference of distinct contexts; rich visual evidence does. Whether biological hippocampi obey the same allocation principle remains an open empirical question.

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix **??**), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix **??**.

5.4 Implications and scope

Connection to information bottleneck theory The capacity-allocation principle is naturally framed as a *task-driven information bottleneck*. The IB principle of **??** characterises representations as trading task-relevant information off against input-redundant detail. Our setting imposes the analogous trade-off under a finite recurrent-memory budget: a policy with bounded $\mathbf{h}_2$ capacity must allocate dimensions between (i) directly carrying encoder-derived visual features and (ii) integrating GPS-sensor history across time. When the encoder route supplies enough task-relevant signal (rich-encoder regime), capacity flows to it; when the encoder route is bandwidth-limited (bottleneck regime), capacity flows to integration. The LOSO scene-invariance result (Figure **??**) gives this IB framing a concrete observable: a bottlenecked encoder, having no scene-specific features to use, is *forced* to encode position invariantly across scenes; a rich encoder is free to encode position scene-conditionally, and policy gradient does so because a scene-conditional code is cheaper to learn given the available features. This connects directly to the invariance–disentanglement reading of the IB (**?**): bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding. We do not *derive* our findings from IB — our setup contains no explicit information-theoretic objective during training — but the observed monotonic anti-correlation between encoder bandwidth and integrated spatial code (Figure **??**) and the LOSO invariance dichotomy (Figure **??**) are the emergent signatures one expects when policy gradient acts as an implicit task-driven IB regulariser on a finite-capacity recurrent state. We use IB descriptively rather than mechanistically: **?** showed several specific claims of IB-deep-learning theory (a distinct compression phase, its causal link to generalisation, its origin in SGD stochasticity) do not hold for ReLU networks, and that MI estimates on deterministic networks depend on implicit binning or noise assumptions. Our MINE estimates (Appendix **??**) are interpretable as lower bounds on $I(\mathbf{h}_2; \text{pos})$ given the statistics-network capacity, and the cross-cond *ordering* of MI is the robust quantity rather than absolute nat values. The IB-invariance reading is a useful analogy that maps onto our LOSO observable; we do not claim a direct causal IB-explains-generalisation chain.

Falsifiable prediction for transformer-based navigators The capacity-allocation principle is architecture-agnostic: it predicts a content-level allocation in any system where a finite-capacity memory module reads a pretrained encoder. For transformer-based PointGoal navigators (e.g. **?**), where "memory" lives in the KV cache or attention over rollout history, the principle predicts the *same* monotonic anti-correlation reported in Figure **??**: attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders. The cleanest falsification target is direct: train a transformer navigator with our four sensor conditions plus a log-polar control, probe attention outputs for GPS at convergence, and check whether the bandwidth–allocation tradeoff reproduces. Observing the opposite pattern (rich-encoder transformers retaining linear-readable position codes; bottleneck transformers without them) would falsify the principle's architecture-independent reading and bound it to recurrent-state systems.

Implications for vision–language and world-model systems The principle has implications for *non-navigation* systems built around rich pretrained visual encoders. Vision–language models that inherit foundation visual encoders are documented to fail at fine-grained spatial reasoning even when surface-level visual recognition is excellent (**?**); one default account is a training-distribution gap, but capacity allocation offers a content-level alternative: a richer encoder front-end may actively divert capacity away from the integration-style spatial code that downstream substrates would otherwise carry. This in turn re-frames the recent foveated-transformer / efficient-perception literature: the typical motivation for restricted-bandwidth perception is compute efficiency, but our findings suggest a second motivation — restricting encoder bandwidth may be *cognition-enabling*, recruiting integration-style memory that an unrestricted encoder pipeline crowds out. Both predictions are testable in current frontier architectures.

The relevant axis appears to be encoder spatial-feature variety per step rather than raw input bandwidth: coarse's 48^2 input with a 1×1 feature map carries a strong linear top-layer GPS code while uniform's 256^2 with 4×4 output does not. A log-polar foveation control (Appendix **??**) targets the same encoder-spatial-output axis with a single intermediate point ($\sim 2\times2$); a finer multi-point input-resolution sweep is left to future work. Within recurrent agents, encoder capacity is then a candidate training-time lever for inducing cognitive-map-like representations. Whether sensor–memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is a separate open question.

5.5 Implications for cognitive neuroscience

The four conditions provide a controlled silicon-side testbed for the comparative cognitive-neuroscience observations our biological precedent (§**??**) draws on, with the encoder, memory, policy, training distribution, and reward held fixed. We do not claim our agent demonstrates the bio-AI parallel; we use it to generate three falsifiable predictions back into the experimental literature.

Hidden-state inference framework, sharper Under **?**'s account, place fields remap when sensory evidence supports distinct hidden states. Our LOSO catastrophe in rich-encoder conditions is exactly the predicted regime: rich per-step visual evidence supports scene-specific hidden states, and the linear position direction reorganises across rooms (analogous to global remapping). Bottleneck conditions lack this evidence and exhibit no remapping (a single scene-invariant code). The capacity-allocation principle adds a missing piece to this framework: *the agent does not freely choose how much sensory variety its encoder provides; that choice is anatomical and species-fixed*. Our results predict that within-species manipulations that change effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision) should shift hippocampal coding along the no-remapping vs. global-remapping axis we exhibit in silicon.

Place-cell vs. view-cell dichotomy The Skaggs analysis (Figure **??**) shows higher place-unit *count* but no per-unit increase in spatial information for foveated vs. blind, consistent with **?**'s primate-fovea $\rightarrow$ view-cell hypothesis. This makes a falsifiable prediction at the implementation level: gaze-fixed primates should show a coding regime closer to rodent place cells than to free-gaze primate view cells, all else equal. The closest existing experiment is **?**'s primate hippocampal recording under restricted gaze; a within-animal contrast between free and restricted gaze would directly test the prediction.

Capacity allocation as a unifying organising principle The Skaggs, LOSO, predictive-horizon, MINE, and subspace-divergence findings cohere under one allocation rather than five separate phenomena (§**??**). Cognitive map theory has long catalogued such phenomena separately (**????**); our setting suggests they may be views of a single mechanism — a finite-capacity recurrent state allocating dimensions between a sensory-history integration route and a perceptual route, with sensor structure setting the allocation point. Whether biological hippocampi obey the same finite-capacity allocation is the central question this silicon model raises but cannot itself answer.

5.6 Limitations

(i) *Single seed per condition*, following the convention of the comparative-cognition modelling literature this paper sits in (**????**). Rigour at the cross-condition level instead derives from convergent multi-method evidence: multiple independent analytical lenses (linear and 2-layer MLP probing, linear CKA, generalised Procrustes shape distance, principal-angle subspace divergence, LOSO scene-invariance, lag-k probe, 5×5 memory transplant, place-cell signature via Skaggs MI, excursion-forgetting) all rank the four conditions in the same direction — a convergence unlikely to be produced by single-seed noise. An independent retraining of the blind condition with a different random seed and effective batch size (Appendix **??**) reproduces the qualitative ranking, providing one cross-setup robustness check. Per-condition seed-variance error bars over a full multi-seed grid are reported as future work. (ii) *Frame budget* is uniformly 250M across all conditions (§**??**); the cross-training probe trajectory (Figure **??**) verifies that the bandwidth–allocation rank ordering emerges before this horizon and is stable across training. (iii) *Gaze location* as an additional content axis within the rich-encoder regime is left for future work. (iv) *Foveation model:* $\sigma_{\max} = 8$ Gaussian blur is one operationalisation; a log-polar foveation control (Appendix **??**) tests whether the H1 mechanism depends on this choice, and a strength sweep over $\sigma_{\max}$ is reported as future work. (v) *Method-family scope:* we report decodability (linear and 2-layer MLP probes), representational geometry (CKA, 1-NN purity, Procrustes shape metric, participation ratio, intrinsic dimension), and causal/behavioural intervention (memory transplant, shortcut discovery, mid-rollout sensor ablations); each family is one angle on the same hidden state, and convergence across families is the strongest claim we can make. Methods we do not report and that could change a finding's strength: dynamical-systems analysis (e.g., fixed-point/slow-point structure (**?**)) which would speak to the format-divergence finding (H2) at the level of computational primitives, and information-theoretic estimators of $I(h_t; \text{position}_{t+k})$ which would replace decodability R^2 with predictive-horizon nats; both are left for follow-up work. (vi) *Architecture scope and the LSTM-artefact concern.* We use a recurrent (LSTM) backbone as a single-architecture case study for direct comparability with the emergent-maps literature (**???**). A reasonable concern is whether the encoder–memory race is an artefact of LSTM-specific limitations (e.g. vanishing gradients across long horizons, restricted gating capacity). Two pieces of within-paper evidence argue against the artefact reading. (a) MLP probes (Appendix **??**) recover position information in rich-encoder conditions at top-layer $R^2 \in [0.51, 0.73]$: the LSTM *does not* discard position; rather the *linear* policy-readable axis tracks encoder bandwidth, a property of the linear DD-PPO action head reading the memory rather than of LSTM gating. (b) Cross-condition memory transplant (§**??**) is a behavioural test outside any probe framework and reproduces the bottleneck-vs-rich-encoder ordering, ruling out probe-specific artefacts. The encoder–memory race is in principle architecture-agnostic — any finite-capacity memory read by a linear policy head should show the same tradeoff — and the most direct generalisation test (training a transformer-based navigator under the same conditions and probing the attention/KV-cache state for position) is the falsification target identified in §**??**; we do not verify it here. (vii) *Algorithm-and-task scope:* DD-PPO + PointGoal in Habitat with ResNet-18 visual encoder. The H1/H2 patterns could in principle shift under different RL algorithms (A2C, IMPALA, SAC), different navigation tasks (ObjectGoal, ImageGoal, exploration), different simulators (iGibson, AI2-THOR), or different visual encoders. Each is a deliberate scope choice for direct comparability with (**?**).

6 Conclusion

Using a recurrent (LSTM) PointNav agent as a controlled silicon model, a four-condition visual-pathway ablation suggests sensor structure as a representational-content axis whose effects cohere under a single capacity-allocation principle. Five concomitant signatures of the same allocation are observed: a magnitude axis (the bandwidth–allocation tradeoff, H1); a format axis (condition-specific representational subspaces with catastrophic cross-condition probe transfer, H2); a scene-invariance axis (bottleneck-induced linear position-axes generalise across rooms while rich-encoder ones do not, H3); a place-cell-like population-coding signature (the bottleneck condition recruits more LSTM units into spatial coding, mirroring sensory-deprivation hippocampal compensation); and a probe-readable vs policy-used dissociation. The bandwidth–allocation ordering is preserved on held-out MP3D, and the pattern is at high level consistent with independent animal-navigation findings on hippocampal compensation under sensory deprivation, cross-modality remapping, and the no-vs.-global remapping axis under hidden-state inference (**?**); we frame the bio–AI connection as an open hypothesis to test rather than as a demonstrated parallel. The cross-condition observations are single-seed per condition, following the comparative-cognition modelling convention (**???**); rigour at the cross-condition level comes from the convergence across multiple independent analytical lenses listed above. An independent-retraining replication of the blind condition reproduces the qualitative ranking. The log-polar foveation control (Appendix **??**) is the falsifiable single-point encoder-spatial-output test. A finer multi-point scaling sweep, a full multi-seed grid for per-condition variance error bars, and gaze-placement controls within the rich-encoder regime are reported as future work. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators or other persistent-state architectures; we do not verify this here.

Code and checkpoints. Code is available at `https://github.com/alunxu/foveated-cog-map`; the four canonical converged checkpoints plus intermediate cross-training checkpoints (used for the §**??** substitution-dynamics figure) are hosted at `https://huggingface.co/alunxu/spatial-memory-checkpoints`.

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   Justification: §3.7 commits to open-source release of the training and probing code. Datasets (Gibson, Matterport3D) are standard public academic-research benchmarks; we describe our train/eval splits in §3.1.

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   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
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6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, Hewitt–Liang control). Appendix A gives the ablation's loss formulation and coefficient.

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7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: We report single-seed numbers per condition. This is documented as Limitation (ii) in §5.4. Multi-seed replication for tighter confidence intervals is in the immediate follow-up.

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   Answer: [Yes]

   Justification: §3.7 reports per-condition training budget (5–7 days on 2×V100 for 250M frames) and probing budget (<1h per condition on one V100).

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: §5.3 discusses implications — sensor design as a tunable axis for interpretable embodied agents. Negative societal impacts are minimal: the work is fundamental research on simulated indoor navigation with no direct deployment pathway.

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A Reproducibility: hyperparameters and implementation notes

Training hyperparameters (DD-PPO, all five conditions identical) 4-GPU `torchrun`; `num_steps=256`, learning rate `lr=2.5e-4` with `lr_decay=False`; `num_environments=16/worker`; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from **?**.

Architectural details (sensor encoding) Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (`nn.Embedding` with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the **?** Wijmans-default sensor stack.

Implementation notes (foveated condition) The foveated condition disables the running-mean-and-variance input normaliser (`RunningMeanAndVar`) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected; the ablation is in the codebase under `src/habitat/foveated_normalised_policy.py` (normaliser *enabled*) for future work that wants to revisit this choice.

Implementation notes (numerical stability) Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) gradient sanitisation in `wijmans_policy._safe_before_step` replaces NaN/Inf gradients with zero before each PPO update; (ii) a forward-looking `geodesic_distance` clamp in `nan_safe_measures.py` prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories.

Log-polar foveation transform The log-polar control (Appendix **??**) uses `LogPolarFoveationTransform` in `src/habitat/torch_foveation.py`: resample the $128{\times}128$ avg-pooled image onto a $64{\times}64$ ($n_\rho \times n_\theta$) log-polar grid before the ResNet-18 backbone. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

B Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

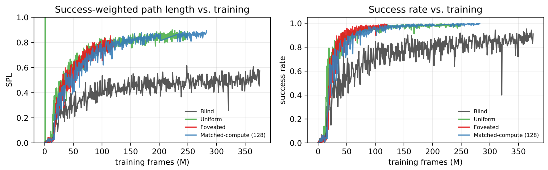
Length-matching ablation. Under deterministic rollouts, episode lengths span ∼100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the H1 ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

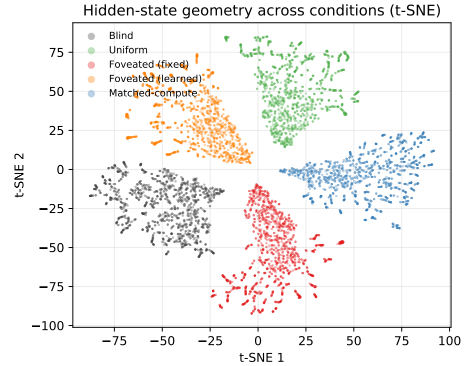
C Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure **??**c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same



(a) Training curves (TensorBoard event-file scalars). All three sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all four conditions (perplexity 30). The four conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§**??**).

Figure 13: Supplementary visualisations referenced from the main text.

5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, uniform 0.55 (vs. Ridge's chance R²). The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the log-polar foveation control (Appendix **??**) targets the encoder-spatial-output axis at one intermediate point.

D Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table **??**); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all four conditions ($R^2 \geq 0.81$), and the real-vs-permuted-label probe gap (Hewitt–Liang selectivity, §**??**) tracks GPS R^2; the chance-level rich-encoder GPS R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix **??**.

Linear-probe artefacts. Two complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the 5×5 matrix's asymmetry adds structure no linear-similarity measure can read. *(b) Non-linear probe:* a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix **??**). H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.

E Supplementary visualisations

E.1 Capacity-allocation: additional geometric tests

The three figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-allocation (Fig. **??**) and LOSO scene-invariance (Fig. **??**) results in §**??**.

E.2 Single-direction β-scrubbing reveals redundant population coding

The Ridge probe identifies a 2-d direction β in $\mathbf{h}_2$ that linearly predicts (x, y). We test whether this direction is *causally responsible* for the recoverable position code by projecting $\mathbf{h}_2$ onto the

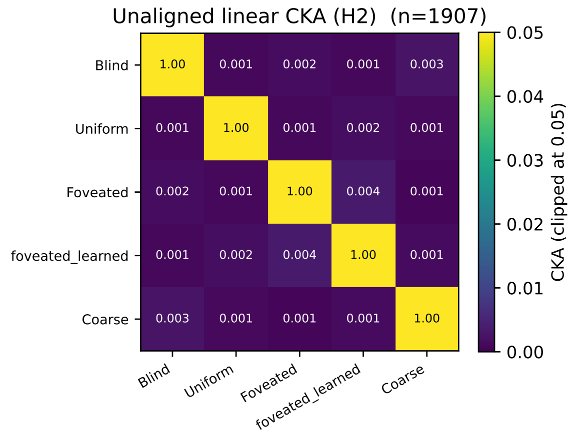


Figure 14: Unaligned linear CKA (?) between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure **??**); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

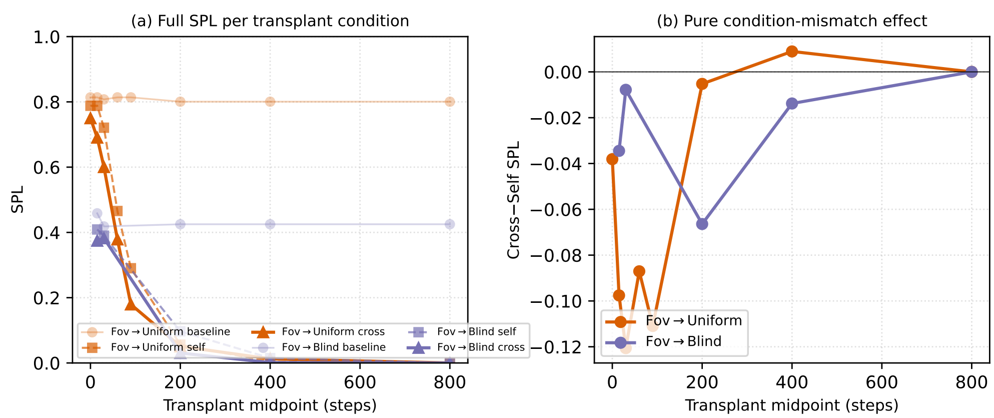


Figure 15: Memory-transplant midpoint sweep on two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps (n=100–150 episodes/cell). The full 4×4 matrix in Figure **??** (right) is at midpoint t=30. *(a):* full SPL per condition — baseline / self / cross. The t=0 self/cross collapse to a protocol-noise floor (so the diagonal in Figure **??** is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. *(b):* cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at t=30 in Figure **??** is therefore neither a t=30-specific artefact nor a late-midpoint inflation.

null-space of β (rank-510 subspace) and re-evaluating both Ridge and MLP probes (Heimersheim & Nanda 2024 noising-style patching: tests whether β was *necessary* for the encoded behavior (?)).

	Linear orig	Linear scrub	MLP orig	MLP scrub
Blind	+0.95	+0.93	+0.94	+0.95
Coarse	+0.72	+0.62	+0.79	+0.84
Uniform	−1.08	−1.46	+0.42	+0.31

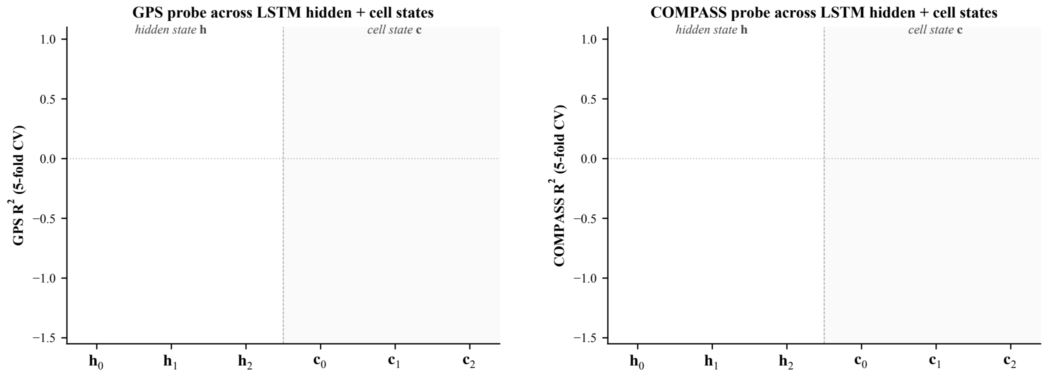


Figure 16: Linear probe (R^2, 5-fold CV) at each LSTM hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left:* GPS. *Right:* compass. Three patterns. *(i) At* $\mathbf{h}_0$ *all conditions decode GPS at* $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). *(ii)* $\mathbf{h}_1$ *dips for every condition* including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for bottleneck. *(iii) Cell states* $\mathbf{c}_\ell$ *do* not *carry the GPS code as cleanly as* $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The H1 claim about "top-layer linearly-decodable GPS" is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

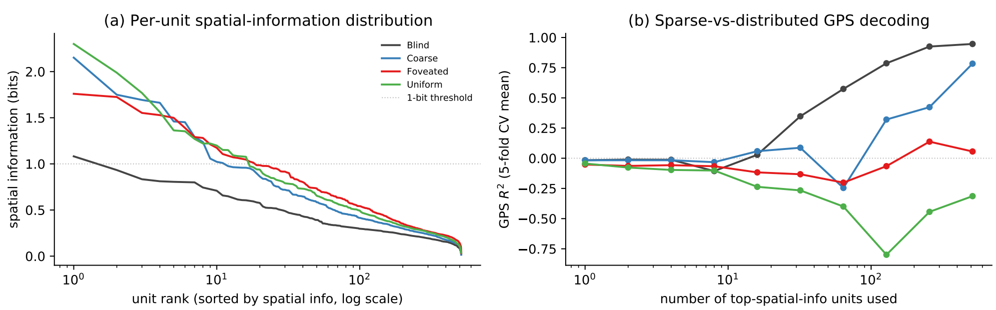


Figure 17: Population coding analysis (referenced from §**??**). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information* $\neq$ *position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

The MLP probe is essentially unaffected by scrubbing across all three conditions ($\Delta R^2 \in [-0.05, +0.11]$). The Ridge probe shows small drops for Blind (-0.02) and Coarse (-0.10), and a more-negative shift for Uniform (the absolute value increases due to the variance of negative R^2 estimates). Two interpretations: (i) the rank-2 subspace identified by Ridge is one of many redundant directions encoding position — when removed, Ridge re-fits to a different direction in the remaining 510 dims; (ii) MLP, having access to non-linear combinations of remaining dims, trivially recovers the position code from this redundant population. This is the population-coding-redundancy property familiar from biological place-cell systems (**?**): single-cell or low-rank ablation does not break the code because of high-dimensional distributed representation. Methodologically, single-direction scrubbing is too weak to causally localise position information in $\mathbf{h}_2$; higher-rank ablation, attribution-based pruning, or path-patching (**?**) would be next steps. Foveated is deferred to the 250M re-probe.

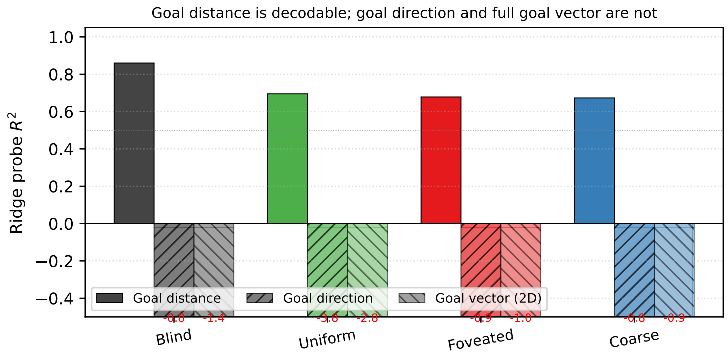


Figure 18: Goal-vector probe R^2 (referenced from §**??**). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

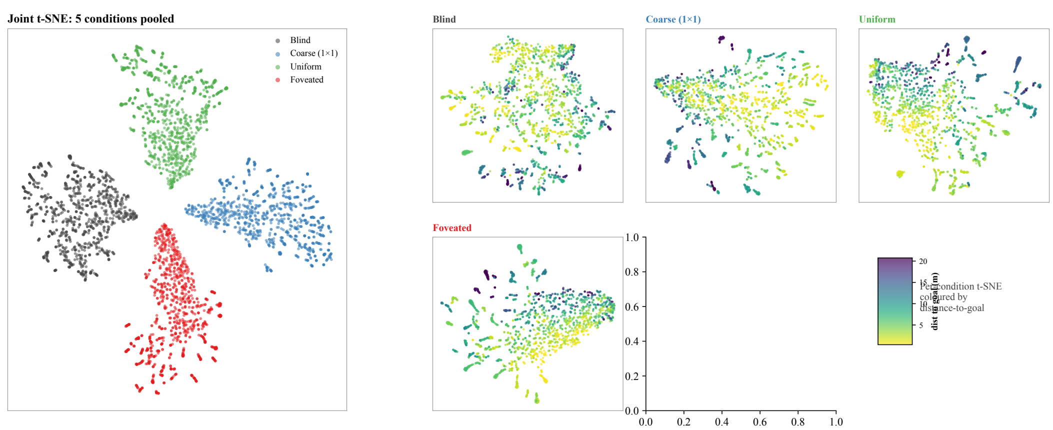


Figure 19: t-SNE of top-layer LSTM hidden states $\mathbf{h}_2$, supplementing the linear-similarity H2 results (§**??**). *Left:* joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right:* per-condition t-SNE coloured by distance-to-goal ($n = 2\,000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind's hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

E.3 Predictive horizon: does h_t encode future positions?

Successor-representation theory (**??**) predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k}. We test the analog claim for our recurrent state: does a linear or MLP probe from $\mathbf{h}_t$ recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

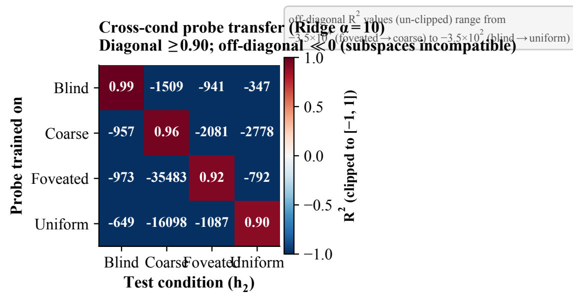


Figure 20: **Cross-condition probe-transfer matrix.** Ridge-probe trained on condition A's top-layer states, applied to condition B's. Diagonal (own-probe) $R^2 \in [0.90, 0.99]$. Off-diagonal R^2 is *catastrophically negative* (-3.5×10^2 to -3.5×10^4 unclipped; clipped to $[-1, 1]$ for visualisation): the position-axes encoded across conditions are not just orthogonal in subspace orientation (Figure **??**) but jointly incompatible at the readout level. Even when the same scenes are visited and the same task is solved, no single linear position-direction transfers between conditions.

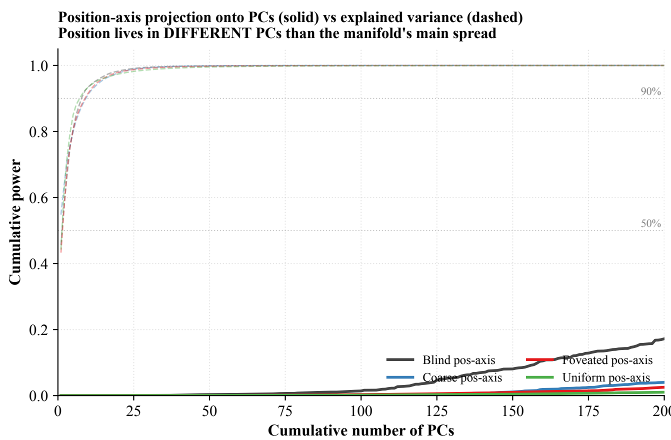


Figure 21: **Position-axis lives in low-variance directions of $\mathbf{h}_2$.** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold's own variance distribution). For all four conditions, the manifold's variance is concentrated in $\sim$ 7–9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of $\mathbf{h}_2$ across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure **??**).

	$k=0$	$k=1$	$k=5$	$k=10$	$k=20$	$k=50$
Linear (Ridge $\alpha=10$) R^2						
Blind	$+0.76$	$+0.77$	$+0.74$	$+0.75$	$+0.72$	$+0.56$
Coarse	$+0.46$	$+0.47$	$+0.47$	$+0.45$	$+0.58$	$+0.21$
Uniform	-1.90	-1.92	-2.08	-2.39	-9.30	-8.49
MLP R^2						
Blind	$+0.88$	$+0.89$	$+0.88$	$+0.73$	$+0.77$	$+0.37$
Coarse	$+0.74$	$+0.74$	$+0.71$	$+0.62$	$+0.67$	$+0.31$
Uniform	-0.43	-0.06	-0.46	-0.34	-0.11	-3.24

Three observations. *(i) Blind exhibits a long predictive horizon.* Linear R^2 from $\mathbf{h}_t$ to $(x, y)_{t+k}$ remains ≥ 0.72 from $k=0$ to $k=20$ (only 0.04-point drop), and 0.56 at $k=50$. That is, the same linear projection that decodes *current* position from $\mathbf{h}_t$ at $R^2 = 0.76$ also decodes positions 20 steps

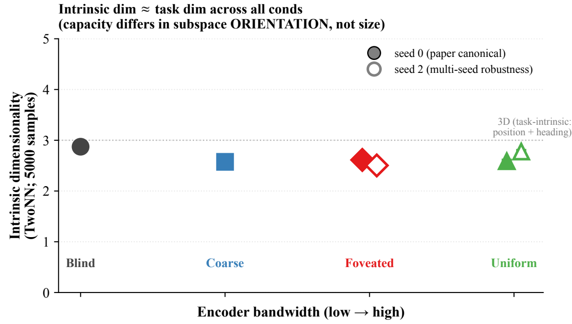


Figure 22: **Intrinsic dimensionality (TwoNN). ?**-style estimate of the intrinsic dimensionality of $\mathbf{h}_2$, on 5 000 samples per condition (paper-canonical seed-0 filled markers; seed-2 robustness check hollow). All four conditions yield ID $\approx$ 2.5–2.9, comparable to the task-intrinsic dimensionality (2-D position + 1-D heading $\approx$ 3). The same intrinsic dim across conditions, combined with strongly differing linear R^2 and orthogonal subspaces (Figure **??**), shows capacity-allocation is about the *orientation* of the task-shaped manifold within the 512-d ambient space rather than about its size.

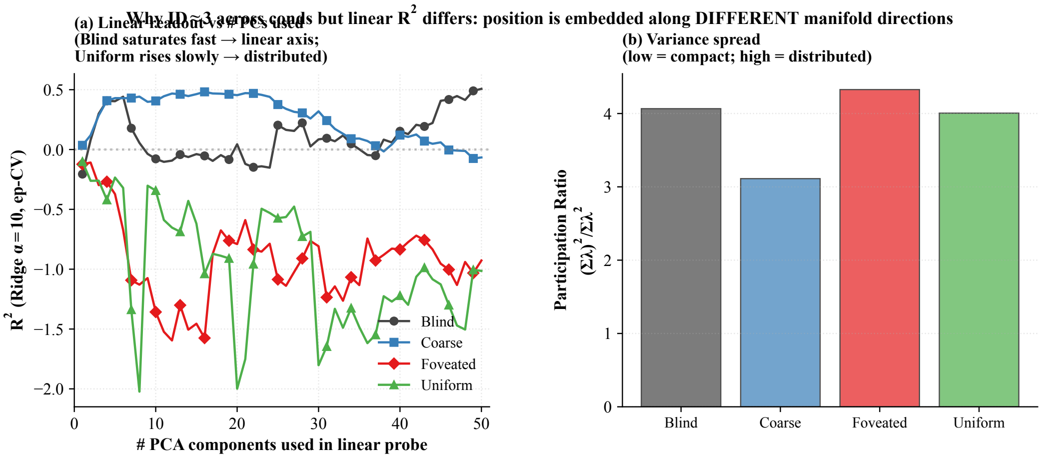


Figure 23: **Linear readout vs. # PCs.** *(a)* Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with $\leq$ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). *(b)* Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure **??**).

in the future at $R^2 = 0.72$. This is a strong form of the predictive-map signature predicted by SR theory: $\mathbf{h}_t$ encodes future-state occupancy, not just current state. *(ii) Coarse follows the same pattern at lower magnitude. (iii) Uniform has no predictive horizon, and indeed no linear current-state code.* Linear R^2 is already negative at $k = 0$ and grows more negative with k. MLP R^2 hovers near zero (≈ 0 to -0.5) before collapsing at $k = 50$ (-3.24). Uniform's $\mathbf{h}_t$ encodes neither current nor future position linearly — consistent with our LOSO scene-invariance reading that uniform's position code is scene-conditional and only non-linearly recoverable, but adds the temporal axis: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation.

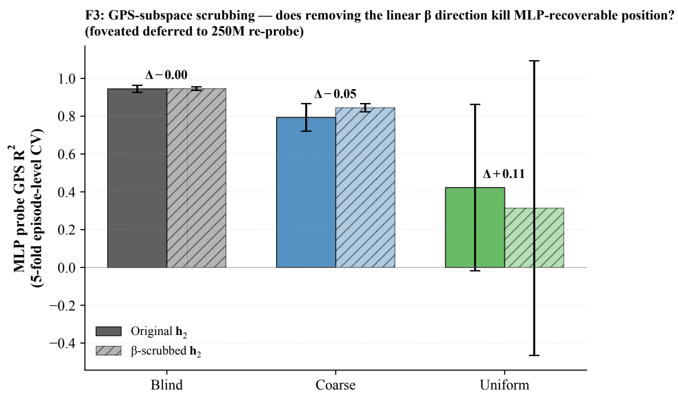


Figure 24: β**-subspace scrubbing.** For each cond (foveated deferred), Ridge β identifies a 2-d direction in $\mathbf{h}_2$ that linearly predicts (x, y); scrubbed $\mathbf{h}_2 = (I - \beta^+\beta)\,\mathbf{h}_2$ removes this 2-d subspace. MLP probe is essentially unaffected by scrubbing across all three conditions: position information is redundantly distributed across many h_2 dimensions, not concentrated in the 2-d β subspace.

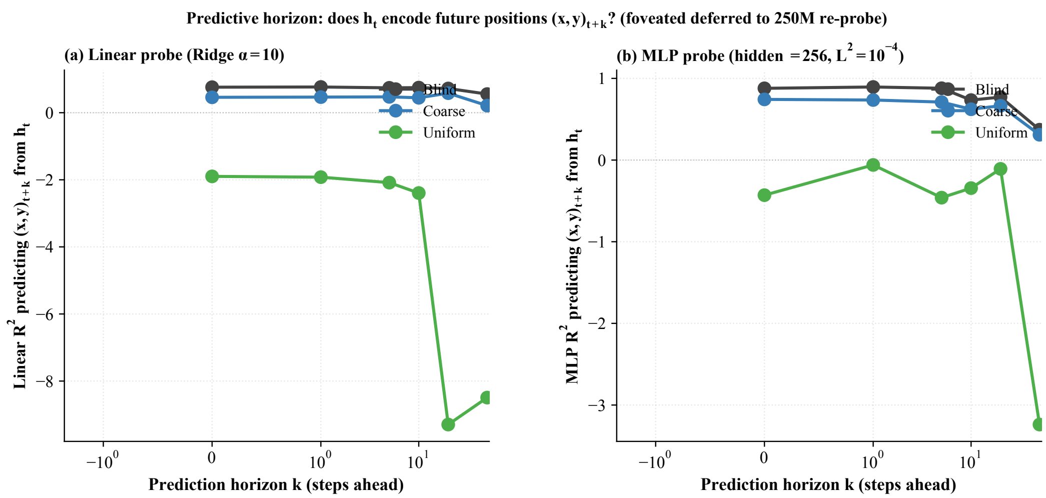


Figure 25: **Predictive horizon: linear and MLP** R^2 **predicting** $(x, y)_{t+k}$ **from** h_t **as a function of horizon** k**.** Blind retains a long predictive horizon (linear R^2 stays ≥ 0.72 to $k = 20$, $+0.56$ at $k = 50$); Coarse a moderate one (linear $R^2 \approx 0.45$ flat to $k = 20$, $+0.21$ at $k = 50$); Uniform has no predictive horizon and no linear current-state code (linear $R^2 < 0$ everywhere; MLP near zero with collapse at $k = 50$). 200 episodes per condition, episode-level 5-fold CV, episode boundaries respected. Foveated is deferred to the 250M-effective re-probe (Phase 3) to avoid mixing the under-converged 174M canonical with the other conditions; we expect the foveated profile to land between Coarse and Uniform.

E.4 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in §**??** on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(\mathbf{h}_2; \mathrm{pos})$ per condition using the Mutual Information Neural Estimator (**?**). MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X; Z) \geq \sup_T \, \mathbb{E}_{P_{XZ}}[T(x, z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x, z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired $(\mathbf{h}_2, \mathrm{pos})$ along episodes; marginal samples are obtained by within-batch shuffling of z. We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4}, 4,000 training steps per condition, and the bias-corrected gradient (eq. 12 of **?**) to mitigate the SGD bias of the log-denominator estimate.

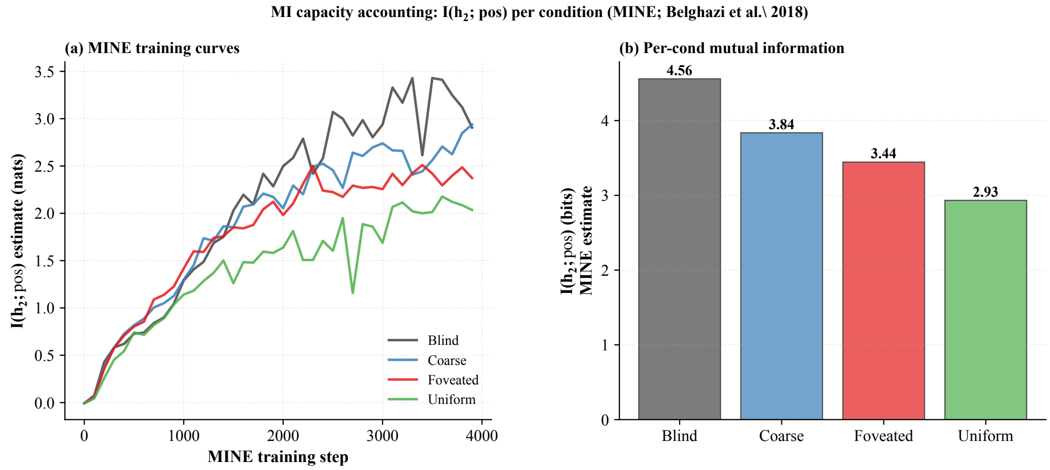


Figure 26: **MINE capacity accounting.** *(a)* MINE training curves: $I_\theta(\mathbf{h}_2; \text{pos})$ in nats vs. optimisation step, per condition. All curves converge by $\sim$ 3,000 steps. *(b)* Final $I(\mathbf{h}_2; \text{pos})$ in bits per condition. Total mutual information decreases monotonically with encoder bandwidth (Blind 4.6, Coarse 3.8, Foveated 3.5, Uniform 2.9 bits), but the decrease is gentle ($\approx 1.5\times$) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The interpretation: rich-encoder conditions retain substantial position information in $\mathbf{h}_2$, but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure **??**).

E.5 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of **?** to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All four conditions show a tight power-law eigenspectrum (Figure **??**, $R^2 > 0.97$). The fitted exponent monotonically tracks encoder bandwidth in a CAP-consistent direction:

Condition	α	R^2	Var. top 10 PCs
Blind	3.92 ± 0.03	0.99	94%
Coarse	4.27 ± 0.04	0.98	91%
Foveated	3.32 ± 0.02	0.99	92%
Uniform	2.85 ± 0.02	0.99	93%

Two observations. (i) All α values are far above Stringer's V1 value ($\alpha \approx 1.04$) and above the critical exponent $1 + 2/d_{\text{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\text{task}} = 3$ stimulus (current position + heading). The recurrent state is therefore much smoother (more compact in its top-PC variance) than the cortical population codes of **?**. We do not claim that Stringer's specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d-dim stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) Rich-encoder conditions have systematically lower α (slower spectral decay) than bottleneck conditions, in the order Coarse $>$ Blind $>$ Foveated $>$ Uniform. Lower α corresponds to variance distributed across more PCs — consistent with the capacity-allocation reading that rich-encoder conditions allocate more recurrent dimensions to visual-feature representation, while bottleneck conditions concentrate variance in fewer top PCs.

E.6 Persistent-memory failure-mode catalog

The main-text Figure **??** shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure **??**) shows four cases per condition spanning the per-condition margin range (most-tries-new $\rightarrow$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5\,\text{m}$ (same-floor old vs. new goal). Within each row,

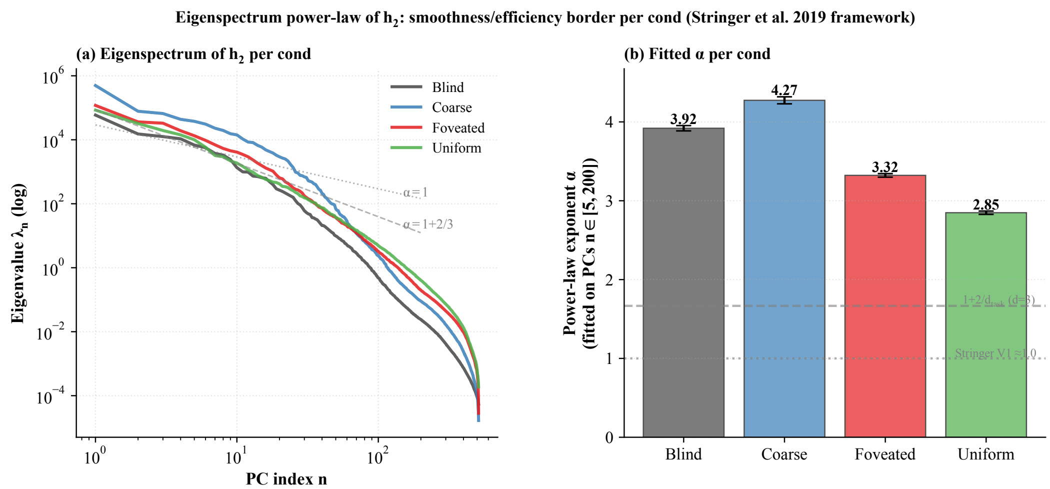


Figure 27: **Eigenspectrum power-law per condition.** *(a)* Eigenvalue λ_n vs. PC index n for $\mathbf{h}_2$, log–log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. *(b)* Fitted power-law exponent α per condition (linear fit on log-λ_n vs. log-n, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.

Table 4: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-4, unified hp)
Encoder feature-map probe	where position information is held	**landed; see §??**

panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

F Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §**??**. All four are submitted; numbers will populate §**??** in revision as they land.

Foveation strength sweep (F1–F4) — planned for follow-up, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 4×4 spatial feature map. A log-polar foveation (**??**) resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2\times2$, much closer to coarse's 1×1 than to uniform's 4×4. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. The preregistered prediction (written before result was available) was that log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.58$ and uniform's ≈ 0, with corresponding behavioural shifts; if log-polar matched uniform, the encoder-spatial-output-dimensionality mechanism would be wrong. The result (next paragraph) falsifies that simple reading.

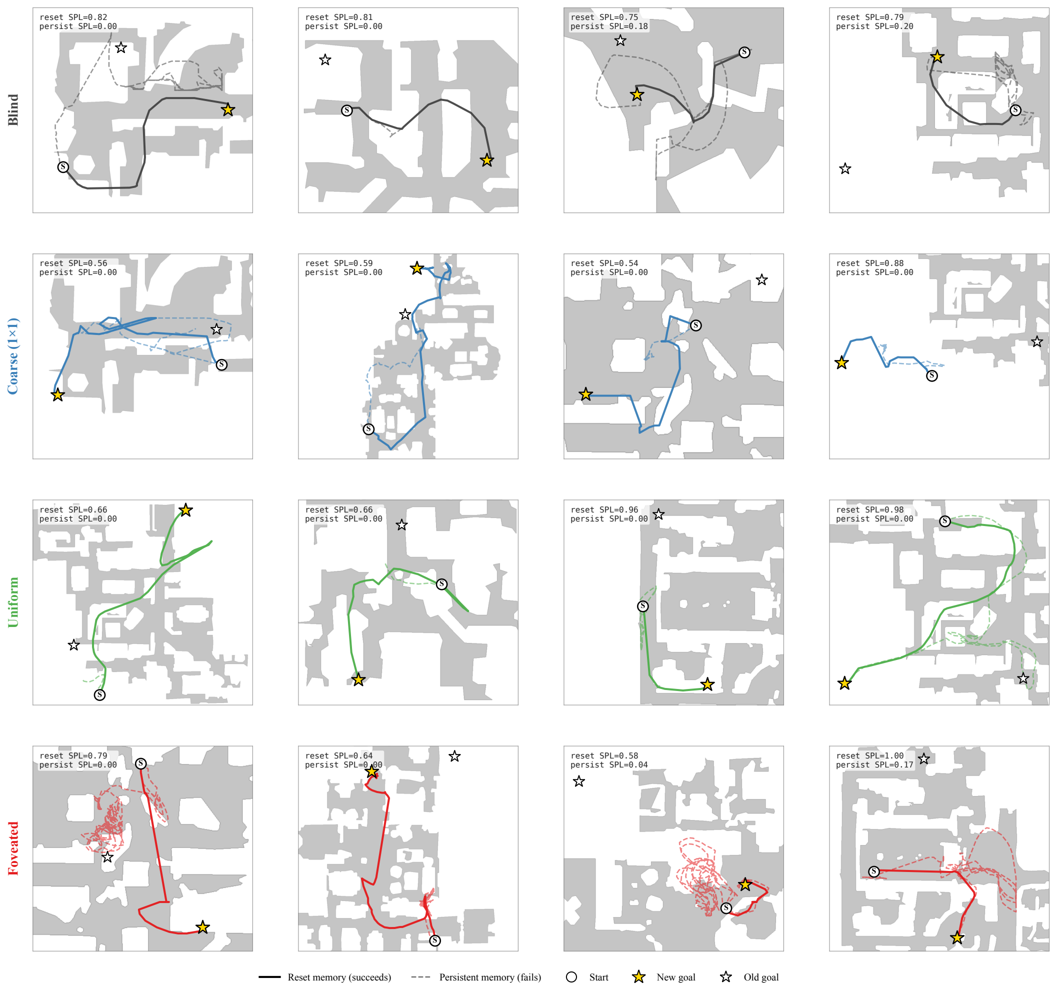
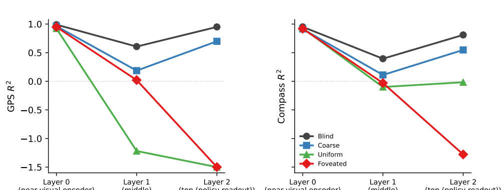
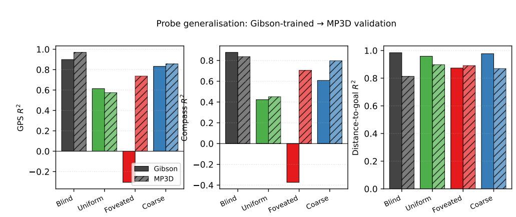


Figure 28: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at "tries-new" (most cases attempt new goal direction but fail to reach); Uniform clusters at "locks-onto-old" (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single seed per condition (cogsci modelling convention; see §**??**).



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all four conditions (single train/test fit; the 5-fold CV mean reported in Table **??** matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure **??**a (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-pass-through partition is the same.

Figure 29: Per-layer and MP3D companion figures (referenced from §**??**).

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists

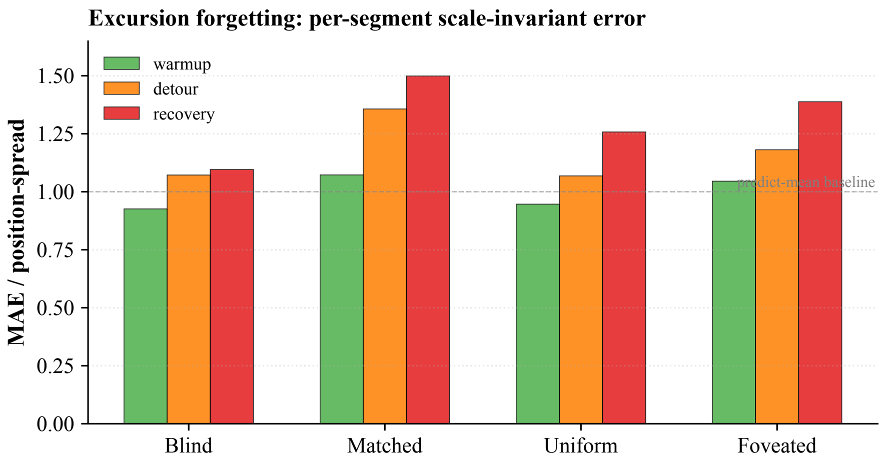


Figure 30: Excursion-forgetting per-segment scale-invariant error (referenced from §**??**). One Ridge probe trained on full-episode pooled hidden states (episode-level 5-fold CV); the same probe tested on each segment of the held-out fold. Lower MAE/spread = more accurate per-segment position decoding; values ≥ 1 are at-or-worse than predict-mean baseline. All four conditions show recovery $>$ warmup error (forgetting); blind's rise (+0.17) is roughly half of the other three (+0.31–0.43), separating its sensor-passthrough regime from the integrated-code regime that coarse and the rich-encoder conditions share. Single seed.

vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper's interpretation.

G Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2\times2$, intermediate between coarse's 1×1 and uniform's 4×4 (verified by direct forward pass through the trained encoder). Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered prediction:** if the H1 mechanism is the simple "encoder spatial-output dimensionality" reading, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse's $+0.58$ than to uniform's ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per step, or another axis. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-49, ~ 245M frames) yields top-layer GPS $R^2 = -0.029 \pm 0.759$ (5-fold CV, $n = 500$ episodes) — *not* in the bottleneck regime, but indistinguishable from foveated ($R^2 = +0.178 \pm 0.659$) and clearly far from coarse ($R^2 = +0.582 \pm 0.099$). The simple "encoder spatial-output dimensionality" framing is therefore *not* sufficient to explain the H1 phenomenon: log-polar's 2×2 output does not place it in the bottleneck regime. We read the result as supporting the alternative "encoder spatial-feature *variety* per step" framing: log-polar's radial sampling preserves rich gaze-centred feature variety per timestep even at low spatial output, so the policy can still source position from the visual route and the LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure **??**) corroborates this reading: log-polar starts at $R^2 = +0.92$ at ~ 50M frames (matching all 4 main conditions' early high R^2), then decays along the rich-encoder trajectory rather than plateauing in the bottleneck range. A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at ~ 250M frames each) was outside this submission's compute window.